

The Two-Variable Factorial Conjecture via Mittag–Leffler Constellations and Fourier–Laplace Rigidity

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Abstract

Let $\mathcal{L}(x^a y^b) = a!b!$ on $\mathbb{C}[x, y]$. We prove the two-variable Factorial Conjecture: if $\mathcal{L}(f^m) = 0$ for every $m \geq 1$, then $f = 0$. The proof has three parts. First, algebraic specialization reduces a hypothetical counterexample to coefficients in $\mathbb{Q}$. Second, an exact Mittag–Leffler generating period rules out every nonflat projective leading form. Its parameterized radial reversion has a computable logarithmic symbol; the positive-interval constellation of Pakovich and Muzychuk forces a primitive Fourier relation, and inverse-branch asymptotics at infinity contradict that relation. In the remaining flat-top case $f_D = (x + y)^D$, the leading singular coefficient of the normalized generating function is

$$\int_0^1 \exp\left(\frac{f_{D-1}(t, 1-t)}{D}\right) dt,$$

so vanishing of all factorial moments forces this integral to vanish. Finally, we prove that for every nonconstant $q \in \overline{\mathbb{Q}}[x]$ and distinct $A, B \in \overline{\mathbb{Q}}$, the integral $\int_A^B e^{q(x)} dx$ is transcendental. The arithmetic proof uses an inhomogeneous E -function moment system, semisimplicity of its Fourier–Laplace connection, horizontal-endomorphism rigidity in the Jacobian algebra of q , Beukers’ refinement of the Siegel–Shidlovskii theorem, and a Cauchy-transform moment argument. Applied to $q(t) = f_{D-1}(t, 1-t)/D$, this gives the final contradiction.

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1 Introduction

For $g \in \mathbb{C}[x, y]$, define the factorial functional by

$$\mathcal{L}(x^a y^b) = a!b!, \quad a, b \geq 0. \quad (1.1)$$

The Factorial Conjecture was introduced by van den Essen, Wright, and Zhao in their study of the Image Conjecture [7, §4]. Its two-variable form is the following.

Theorem 1.1 (Two-variable Factorial Conjecture). *For every $f \in \mathbb{C}[x, y]$,*

$$\mathcal{L}(f^m) = 0 \quad \text{for all } m \geq 1 \quad \implies \quad f = 0.$$

The homogeneous two-variable case was proved by Liu and Sun [5]. The inhomogeneous problem is not formally reduced to that case because radial integration couples the homogeneous layers through Gamma factors.

Write a nonzero polynomial of exact degree D as

$$f = f_D + f_{D-1} + \cdots + f_0,$$

where every f_j is homogeneous of degree j , and put

$$A_j(t) = f_{D-j}(t, 1-t), \quad 0 \leq j \leq D. \quad (1.2)$$

The proof of Theorem 1.1 follows the implication chain

$$\begin{aligned}
& \text{counterexample over } \mathbb{C} \implies \text{counterexample over } \overline{\mathbb{Q}}, \\
& \implies A_0(t) \text{ is constant,} \\
& \implies f_D = (x + y)^D \text{ after scaling,} \\
& \implies \int_0^1 e^{A_1(t)/D} dt = 0, \\
& \implies \text{contradiction.}
\end{aligned}$$

The second implication is the nonflat-top theorem proved in Sections 3 to 7. The fourth is the flat first-symbol calculation in Section 8. The final contradiction is supplied by the algebraic exponential-integral theorem of Theorem 9.1, proved in Sections 10 to 16.

2 Algebraic specialization

Proposition 2.1 (Algebraic specialization). *If a nonzero counterexample to Theorem 1.1 exists in $\mathbb{C}[x, y]$, then there is one in $\overline{\mathbb{Q}}[x, y]$ with the same monomial support.*

Proof. Let

$$f = \sum_{\alpha \in S} c_\alpha x^{\alpha_1} y^{\alpha_2}$$

be a nonzero counterexample with finite support S , and choose $\alpha_0 \in S$. Replacing f by $c_{\alpha_0}^{-1} f$ preserves every moment vanishing, since

$$\mathcal{L}((c_{\alpha_0}^{-1} f)^n) = c_{\alpha_0}^{-n} \mathcal{L}(f^n) = 0 \quad (n \geq 1),$$

and it preserves the support. We may therefore impose the normalization $c_{\alpha_0} = 1$.

For every $n \geq 1$, the quantity $\mathcal{L}(f^n)$ is a polynomial with integer coefficients in the remaining coefficients c_α : the multinomial coefficients are integers, and applying $\mathcal{L}$ to each resulting monomial multiplies its coefficient by two factorials. Put

$$R = \overline{\mathbb{Q}}[c_\alpha, u_\alpha : \alpha \in S \setminus \{\alpha_0\}],$$

and let $I \subseteq R$ be generated by all moment polynomials $\mathcal{L}(f^n)$, $n \geq 1$, together with the Rabinowitsch equations

$$u_\alpha c_\alpha - 1 = 0 \quad (\alpha \in S \setminus \{\alpha_0\}).$$

Although the displayed family of moment generators is infinite, the ideal is proper: evaluation at the original normalized complex coefficient vector, with $u_\alpha = c_\alpha^{-1}$, annihilates every generator. Hence I is contained in a maximal ideal $\mathfrak{m}$ of R . By the weak Nullstellensatz, $\mathfrak{m}$ is the vanishing ideal of a point with coordinates in $\overline{\mathbb{Q}}$. At that point the Rabinowitsch equations force every coefficient indexed by S to remain nonzero, while all the factorial-moment polynomials vanish. The resulting polynomial in $\overline{\mathbb{Q}}[x, y]$ therefore has support exactly S and satisfies every factorial-moment equation. $\square$

3 Radial–projective coordinates

Lemma 3.1 (Gamma integral). *For every $g \in \mathbb{C}[x, y]$,*

$$\mathcal{L}(g) = \int_0^\infty \int_0^\infty g(x, y) e^{-x-y} dx dy. \quad (3.1)$$

Proof. The integral is absolutely convergent for every polynomial, since a polynomial has at most polynomial growth and the weight is e^{-x-y} . It is therefore enough by linearity to test the monomial $x^a y^b$. Fubini's theorem and the Gamma integral give

$$\int_0^\infty \int_0^\infty x^a y^b e^{-x-y} dx dy = \Gamma(a+1)\Gamma(b+1) = a!b!.$$

□

Let f have exact degree D , with homogeneous decomposition and projective layers A_j as in (1.2).

Lemma 3.2 (Projective degree and the factor $x + y$). *Let f_D be homogeneous of degree D , and write*

$$f_D = (x + y)^m g_D, \quad x + y \nmid g_D.$$

Then

$$\deg f_D(t, 1 - t) = D - m.$$

In particular, $f_D(t, 1 - t)$ is constant if and only if f_D is a scalar multiple of $(x + y)^D$.

Proof. Since $t + (1 - t) = 1$,

$$f_D(t, 1 - t) = g_D(t, 1 - t).$$

The polynomial on the right has degree at most $D - m$. Its coefficient of t^{D-m} is $g_D(1, -1)$. For a homogeneous polynomial g_D , the equality $g_D(1, -1) = 0$ is equivalent to the identity $g_D(x, -x) = 0$, and hence to the divisibility $x + y \mid g_D$. The defining assumption therefore gives $g_D(1, -1) \neq 0$, so the degree is exactly $D - m$. The final assertion follows by taking $D - m = 0$. □

For $r > 0$, homogeneity gives

$$f(rt, r(1 - t)) = r^D \left(A_0(t) + \frac{A_1(t)}{r} + \cdots + \frac{A_D(t)}{r^D} \right). \quad (3.2)$$

Proposition 3.3 (Exact radial–projective formula). *For every $n \geq 0$,*

$$\begin{aligned} I_n(f) &:= \frac{\mathcal{L}(f^n)}{\Gamma(Dn + 2)} \\ &= \frac{1}{\Gamma(Dn + 2)} \int_0^1 \int_0^\infty e^{-r} r f(rt, r(1 - t))^n dr dt \\ &= \frac{1}{\Gamma(Dn + 2)} \int_0^1 \int_0^\infty e^{-r} r^{Dn+1} \left(A_0(t) + \frac{A_1(t)}{r} + \cdots + \frac{A_D(t)}{r^D} \right)^n dr dt. \end{aligned} \quad (3.3)$$

In the last integral the apparent singularity at $r = 0$ is removable: the full integrand is the preceding nonsingular expression $e^{-r} r f(rt, r(1 - t))^n$.

Proof. Use the change of variables

$$x = rt, \quad y = r(1 - t), \quad r \geq 0, \quad 0 \leq t \leq 1.$$

The absolute value of its Jacobian determinant is r , and $x + y = r$. Substitution into (3.1) gives the first integral in (3.3). For $r > 0$, insert (3.2); the resulting equality extends across $r = 0$ by the removable-singularity observation in the statement. $\square$

4 Mittag–Leffler generating periods

Throughout Sections 4 to 7, fix $D \geq 1$ and polynomials

$$A_0, A_1, \dots, A_D \in \mathbb{C}[t].$$

Put

$$P(t, r) = \sum_{j=0}^D A_j(t) r^{D-j}. \quad (4.1)$$

For the projective layers of a degree- D polynomial, this is exactly $f(rt, r(1 - t))$.

For $\beta \in \mathbb{N}$, define the Mittag–Leffler function

$$E_{D,\beta}(z) = \sum_{n \geq 0} \frac{z^n}{\Gamma(Dn + \beta)}. \quad (4.2)$$

Lemma 4.1 (Uniform Mittag–Leffler growth and initial holomorphy). *For $\beta \in \{2, 3\}$ there are constants $C, N > 0$, depending only on D and β , such that*

$$|E_{D,\beta}(z)| \leq C(1 + |z|)^N \exp(C|z|^{1/D}) \quad (z \in \mathbb{C}). \quad (4.3)$$

Let $K \subset \mathbb{C}$ be compact. Then there is $C_K > 0$ such that

$$|P(t, r)|^{1/D} \leq C_K(1 + r) \quad (t \in K, \quad r \geq 0). \quad (4.4)$$

Consequently, for $|s|$ sufficiently small, the integrals in (4.8) and (4.9) converge absolutely and locally uniformly in t on K and in s . They define holomorphic functions of s , and of t whenever t is retained as a complex parameter; termwise integration of the Mittag–Leffler series is valid.

Proof. For $|z| \leq 1$, the entire functions $E_{D,\beta}$ are bounded. If $|z| \geq 1$, choose X with $X^D = z$. The elementary root-of-unity filter, recorded precisely in Lemma 4.4, writes $E_{D,\beta}(X^D)$ as a finite sum of terms of the form $X^{1-\beta} e^{\omega^k X}$ together with a finite Laurent correction. Since $|X| = |z|^{1/D}$, this gives (4.3) after enlarging C and N .

Because $P(t, r)$ has degree at most D in r and its coefficients are bounded on K , one has $|P(t, r)| \leq C_K^D(1 + r)^D$, which is (4.4). Hence the absolute value of either integrand is bounded by

$$C' e^{-r} r(1 + r)^{N'} \exp(C' C_K |s|^{1/D} (1 + r)).$$

For $|s|$ small enough this is dominated by $C'' e^{-r/2} (1 + r)^{N'+1}$, uniformly for $t \in K$. Dominated convergence and the Weierstrass theorem now give all asserted holomorphy and termwise-integration statements. $\square$

Definition 4.2 (Normalized generating periods). Let $I_n = I_n(f)$ denote the normalized radial moment in (3.3). Define

$$\mathcal{G}(s) = \sum_{n \geq 0} I_n s^n, \quad (4.5)$$

$$\mathcal{J}(s) = \sum_{n \geq 0} \frac{I_n}{Dn+2} s^n, \quad (4.6)$$

$$\mathcal{K}(w) = \frac{1}{w} \mathcal{J}(w^{-1}). \quad (4.7)$$

Proposition 4.3 (Exact Mittag-Leffler representations). For $|s|$ sufficiently small,

$$\mathcal{G}(s) = \int_0^1 \int_0^\infty e^{-r} r E_{D,2}(sP(t,r)) dr dt, \quad (4.8)$$

$$\mathcal{J}(s) = \int_0^1 \int_0^\infty e^{-r} r E_{D,3}(sP(t,r)) dr dt. \quad (4.9)$$

Consequently, for $|w|$ sufficiently large,

$$\mathcal{K}(w) = \int_0^1 \mathcal{B}(w,t) dt, \quad \mathcal{B}(w,t) = \frac{1}{w} \int_0^\infty e^{-r} r E_{D,3}\left(\frac{P(t,r)}{w}\right) dr. \quad (4.10)$$

Proof. Expand the Mittag-Leffler functions and use (3.3). The locally uniform integrable majorant from Lemma 4.1 justifies termwise integration and proves both identities. Substituting $s = w^{-1}$ and multiplying by w^{-1} gives (4.10). $\square$

Lemma 4.4 (Root-of-unity filter with low-degree corrections). Let $m \geq 0$, let $\omega = e^{2\pi i/D}$, and define the finite Laurent polynomial

$$C_{D,m}(X) = \sum_{\substack{0 \leq \ell < m \\ \ell \equiv m \pmod{D}}} \frac{X^{\ell-m}}{\ell!}.$$

Then

$$E_{D,m+1}(X^D) = \frac{1}{DX^m} \sum_{k=0}^{D-1} \omega^{-mk} e^{\omega^k X} - C_{D,m}(X). \quad (4.11)$$

In particular, $C_{D,1} = 0$ unless $D = 1$, while $C_{D,2} = 0$ for $D \geq 3$, $C_{2,2}(X) = X^{-2}$, and

$$C_{1,2}(X) = X^{-2} + X^{-1}.$$

Proof. Expand the exponentials. The root-of-unity sum retains precisely the indices $\ell \equiv m \pmod{D}$:

$$\frac{1}{DX^m} \sum_{k=0}^{D-1} \omega^{-mk} e^{\omega^k X} = \sum_{\substack{\ell \geq 0 \\ \ell \equiv m \pmod{D}}} \frac{X^{\ell-m}}{\ell!}.$$

The terms with $\ell \geq m$ form $E_{D,m+1}(X^D)$; the finitely many terms with $\ell < m$ form $C_{D,m}$. $\square$

5 Parameterized radial reversion and the local logarithmic symbol

The local singularity of (4.10) is controlled entirely by the behavior of $P(t, r)$ at $r = \infty$.

Lemma 5.1 (Uniform radial reversion near a nonzero leading value). *Let $t_0 \in \mathbb{C}$ satisfy $u_0 = A_0(t_0) \neq 0$. After shrinking a neighborhood U of t_0 , there is a common exterior region in which*

$$\Phi_t(r) = \left(\frac{P(t, r)}{A_0(t)} \right)^{1/D} = r + \frac{A_1(t)}{DA_0(t)} + O(r^{-1}) \quad (5.1)$$

has a holomorphic inverse

$$r = \chi(t, Y) = Y - \frac{A_1(t)}{DA_0(t)} + O(Y^{-1}). \quad (5.2)$$

All expansions are locally uniform in $t \in U$, and their coefficients are holomorphic in t .

Proof. The quotient $P(t, r)/(A_0(t)r^D)$ is $1 + O(r^{-1})$ uniformly on compact subsets of U . Choose the D -th root asymptotic to 1, giving (5.1). In the coordinate r^{-1} , the derivative of Φ_t at infinity is 1, uniformly in t . The parameter-dependent holomorphic inverse-function theorem gives a common inverse and the expansion (5.2). $\square$

Lemma 5.2 (Uniform exterior-ray deformation). *Under the hypotheses of Lemma 5.1, after shrinking U and increasing R , there are a common exterior sector $\mathfrak{S}$, a point $Y_0 \in \mathfrak{S}$, and a ray*

$$\ell = Y_0 + e^{i\theta}[0, \infty) \subset \mathfrak{S}$$

with the following property. For every $t \in U$, the image curve

$$\Gamma_t = \Phi_t([R, \infty))$$

lies in $\mathfrak{S}$, remains in the common domain of the inverse $\chi(t, Y)$, and is asymptotic to ℓ .

Let $F(t, Y)$ be jointly holomorphic on $U \times \mathfrak{S}$ and satisfy $F(t, Y) = O(Y^{-1})$ there, locally uniformly in t . For η in a sufficiently small sector such that

$$\Re(\eta e^{i\theta}) \geq c|\eta|$$

for some $c > 0$, the difference

$$\int_{\Gamma_t} e^{-\eta Y} F(t, Y) dY - \int_{\ell} e^{-\eta Y} F(t, Y) dY \quad (5.3)$$

extends jointly holomorphically to $(t, \eta) \in U \times \{|\eta| < \varepsilon\}$.

Proof. The expansions in Lemma 5.1 are uniform in t . Thus, after increasing R , the curves Γ_t lie in one narrow exterior sector and admit compatible parameterizations over the ray ℓ for which the transverse displacement is $O(1)$ and the tangent difference is $O(|Y|^{-2})$, locally uniformly in t . Shrinking U if necessary, join the initial point of each Γ_t to Y_0 by a path in a fixed compact annular subset of $\mathfrak{S}$, depending holomorphically on t .

Truncate Γ_t and ℓ at the same large ray parameter M and join the corresponding outer endpoints inside $\mathfrak{S}$. Their distance, and hence the length of this outer connector, is $O(1)$ rather than $O(M)$. On the connector one has $F(t, Y) = O(M^{-1})$. Moreover, after narrowing the η -sector

and the exterior sector, $e^{-\eta Y}$ is uniformly bounded there on every closed admissible subsector together with its vertex $\eta = 0$. The outer-connector integral is therefore $O(M^{-1})$ uniformly on compact subsets of U and of such a closed subsector, and tends to zero.

Cauchy's theorem now identifies (5.3), throughout the admissible sector, with the finite initial-connector integral. That integral is entire in η and holomorphic in t , and therefore furnishes the asserted joint extension to a full disk about $\eta = 0$. The same argument applies after termwise differentiation on smaller compact sets. $\square$

Lemma 5.3 (A parameterized Laplace logarithm). *Let $q_0(t)$ be holomorphic on a neighborhood U , and let $H(t, z)$ be holomorphic for $(t, z) \in U \times \{|z| < \rho\}$. On a fixed sector*

$$0 < |\eta| < \varepsilon, \quad |\arg \eta - \theta| < \frac{\pi}{2} - \delta,$$

consider

$$L(t, \eta) = \int_{Y_0}^{\infty e^{i\theta}} e^{-\eta Y} \left(\frac{q_0(t)}{Y} + \frac{H(t, Y^{-1})}{Y^2} \right) dY.$$

Then

$$L(t, \eta) = -q_0(t) \log \eta + H_0(t, \eta) + \eta \log \eta H_1(t, \eta), \quad (5.4)$$

where H_0, H_1 are jointly holomorphic in t and η near $U \times \{0\}$.

Proof. The Y^{-1} term gives the exponential integral and hence $-\log \eta$ modulo a holomorphic function. Expand

$$H(t, z) = \sum_{m \geq 0} h_m(t) z^m$$

normally on a smaller bidisk. The Laplace transform of Y^{-m-2} is, modulo a holomorphic function, a nonzero scalar multiple of $\eta^{m+1} \log \eta$. Normal convergence, uniform on compact subsets of U , permits termwise integration and gives (5.4). Equivalently, one may subtract finitely many terms and integrate the analytic remainder repeatedly by parts; the Cauchy estimates for H give the required uniform bounds. $\square$

Theorem 5.4 (Parameterized local logarithmic symbol). *Let $t_0 \in \mathbb{C}$ satisfy $u_0 = A_0(t_0) \neq 0$. The germ (4.10), initially defined for large w , admits multivalued analytic continuation to a neighborhood of $(w, t) = (u_0, t_0)$ outside the divisor $w = A_0(t)$. On a local slit neighborhood,*

$$\begin{aligned} \mathcal{B}(w, t) = & -\frac{1}{DA_0(t)} \exp\left(\frac{A_1(t)}{DA_0(t)}\right) \log(w - A_0(t)) \\ & + H_0(w, t) + (w - A_0(t)) \log(w - A_0(t)) H_1(w, t), \end{aligned} \quad (5.5)$$

where H_0, H_1 are jointly holomorphic. Thus the lowest logarithmic symbol depends only on A_0 and A_1 .

Proof. Shrink the neighborhood U from Lemma 5.1 so that A_0 never vanishes. Split the radial integral in (4.10) at a sufficiently large R , increasing R so that $\Phi_t(r)$ is uniformly bounded away from zero for $t \in U$ and $r \geq R$. The compact part $0 \leq r \leq R$ is holomorphic for w near $u_0 \neq 0$ because $E_{D,3}$ is entire.

Choose the branch

$$\lambda = \left(\frac{A_0(t)}{w} \right)^{1/D}$$

near $\lambda = 1$. Since

$$\frac{P(t, r)}{w} = (\lambda \Phi_t(r))^D,$$

Lemma 4.4 with $m = 2$ gives, on the tail,

$$\mathcal{B}(w, t) = \frac{\lambda^{D-2}}{DA_0(t)} \sum_{k=0}^{D-1} \omega^{-2k} \int_R^\infty e^{-r+\omega^k \lambda \Phi_t(r)} \frac{r}{\Phi_t(r)^2} dr + H_{\text{corr}}(w, t), \quad (5.6)$$

where the finite correction H_{corr} is holomorphic near (u_0, t_0) . Indeed, the root filter is used only on the radial tail, where $\lambda \Phi_t(r)$ stays away from zero; hence the negative powers occurring for $D = 1, 2$ have holomorphic, exponentially integrable coefficients. For $k \neq 0$, one has

$$\Re(1 - \omega^k \lambda) > 0$$

uniformly after shrinking the neighborhood, so those tail integrals are holomorphic. Only $k = 0$ can be singular.

In the principal term put $Y = \Phi_t(r)$ and $r = \chi(t, Y)$. Its amplitude becomes

$$e^{Y-\chi(t, Y)} \frac{\chi(t, Y)}{Y^2} \partial_Y \chi(t, Y) = \frac{q_0(t)}{Y} + \frac{H(t, Y^{-1})}{Y^2}, \quad (5.7)$$

where

$$q_0(t) = \exp\left(\frac{A_1(t)}{DA_0(t)}\right)$$

and H is jointly holomorphic. Apply Lemma 5.2 to the amplitude in (5.7). Replacing the image curves $\Phi_t([R, \infty))$ by one fixed ray changes the principal tail only by a function jointly holomorphic in (t, η) at $\eta = 0$.

Put $\eta = 1 - \lambda$. By Lemma 5.3, the principal tail is

$$\frac{\lambda^{D-2}}{DA_0(t)} \left[-q_0(t) \log \eta + \tilde{H}_0(t, \eta) + \eta \log \eta \tilde{H}_1(t, \eta) \right].$$

Finally,

$$\eta = 1 - \left(\frac{A_0(t)}{w}\right)^{1/D} = \frac{w - A_0(t)}{DA_0(t)} + O((w - A_0(t))^2).$$

Hence $\log \eta = \log(w - A_0(t))$ modulo a jointly holomorphic function, and $\eta \log \eta$ is $(w - A_0(t)) \log(w - A_0(t))$ times a holomorphic unit, modulo a holomorphic function. Evaluating the analytic prefactor at $\lambda = 1$ gives (5.5).

The lowest logarithmic coefficient is independent of the local D -th-root sheet. Indeed, suppose a continued branch satisfies $\lambda \rightarrow \omega^q$ as $w \rightarrow A_0(t)$. The unique potentially singular root-filter summand is then $k \equiv -q \pmod{D}$. With $\mu = \omega^k \lambda \rightarrow 1$, the same Laplace calculation applies with $\eta = 1 - \mu$, while its analytic prefactor has limiting root-of-unity factor

$$\lambda^{D-2} \omega^{-2k} \rightarrow \omega^{q(D-2)} \omega^{2q} = \omega^{qD} = 1.$$

Thus every continuation sheet has the coefficient displayed in (5.5). This sheet-independence will be used in the constellation argument. $\square$

Lemma 5.5 (Continuation away from the leading divisor). *Let U be a sufficiently small neighborhood on which A_0 is nowhere zero and the uniform reversion of Lemma 5.1 is valid. Put*

$$\mathfrak{X}_U = \{(w, t) \in \mathbb{C}^\times \times U : w \neq A_0(t)\}.$$

The germ $\mathcal{B}(w, t)$ defined for large w admits joint multivalued analytic continuation along every path in $\mathfrak{X}_U$. Every continued branch is locally holomorphic on $\mathfrak{X}_U$.

Proof. Split the radial integral at a sufficiently large R . On $0 \leq r \leq R$,

$$\frac{1}{w} e^{-r} r E_{D,3} \left(\frac{P(t, r)}{w} \right)$$

is jointly holomorphic for $w \neq 0$, and compact integration preserves this holomorphy.

For the tail, choose locally a branch

$$\lambda^D = \frac{A_0(t)}{w}.$$

After the root-of-unity filter and the change of variable $Y = \Phi_t(r)$, each exponential summand is, up to a locally holomorphic prefactor, of the form

$$L_k(t, \lambda) = \int_{\Gamma_t} e^{-\eta_k Y} F(t, Y) dY, \quad \eta_k = 1 - \omega^k \lambda, \quad (5.8)$$

where

$$F(t, Y) = e^{Y - \chi(t, Y)} \frac{\chi(t, Y)}{Y^2} \partial_Y \chi(t, Y) = O(Y^{-1})$$

is jointly holomorphic on a full common exterior neighborhood.

Fix a point (t_*, λ_*) at which $\eta_k \neq 0$. Choose a ray direction θ and a neighborhood W of (t_*, λ_*) such that

$$\Re(\eta_k e^{i\theta}) \geq \delta > 0 \quad \text{on } W.$$

Inside the common exterior domain, join the initial point of Γ_t to a fixed sufficiently distant ray in direction θ by a finite connector, chosen locally holomorphically in t . The sum of the connector integral and the integral along that ray converges normally and is jointly holomorphic on W .

Whenever two such ray representations have a common decay region, Cauchy's theorem on the truncated sector between the rays identifies them: the outer circular arc tends to zero because of exponential decay and $F(t, Y) = O(Y^{-1})$. The admissible half-planes obtained by varying θ cover $\eta_k \neq 0$, and along any path avoiding 0 one may choose a finite chain of overlapping half-planes. Starting with the original decay region of (5.8), the corresponding ray representations therefore give its analytic continuation along the path. Uniqueness on the overlaps gives the asserted multivalued continuation.

A singularity of the k th summand can therefore occur only when

$$\eta_k = 0 \iff \omega^k \lambda = 1 \iff w = A_0(t).$$

The algebraic parameter λ can branch only around $w = 0$, which is already removed from $\mathfrak{X}_U$.

It remains to check the finite root-filter corrections. For $D \geq 3$ there are none. For $D = 2$ the correction reduces on the tail to an exponentially weighted multiple of $1/P(t, r)$, and for $D = 1$ to a combination of $1/P(t, r)$ and $w/P(t, r)^2$. After increasing R , the polynomial $P(t, r)$ is uniformly nonzero there, so these correction integrals are jointly holomorphic. This proves the asserted continuation and local holomorphy. $\square$

Remark 5.6 (Why the lower layers do not enter the symbol). The degree- $(D - 1)$ layer changes the constant term in the reverted coordinate and exponentiates to the factor in (5.5). The layers $A_2, \dots, A_D$ first occur in the Y^{-2} and lower terms of (5.7); their Laplace transforms therefore begin at order $(w - A_0) \log(w - A_0)$.

6 Positive-interval constellations

Let

$$A = A_0 \in \mathbb{C}[t], \quad d = \deg A \geq 1.$$

Choose a sufficiently large regular value c of A . Join c by pairwise disjoint arcs to every finite critical value of A , and also to $A(0)$ and $A(1)$ if either is not already a critical value. Let S be the resulting star. The preimage $A^{-1}(S)$ is a planar tree, and there is a unique oriented path $\mu_{0,1}$ from 0 to 1.

Let

$$t_1(u), \dots, t_d(u)$$

be the inverse branches of A near c . For each arm γ_s of the star, the path $\mu_{0,1}$ determines an incidence vector

$$f_s = (f_{s,1}, \dots, f_{s,d}) \in \mathbb{Z}^d.$$

Let G_A be the monodromy group and define

$$M_{A,0,1} = \text{span}_{\mathbb{Q}}\{\sigma f_s : \sigma \in G_A, s\}. \quad (6.1)$$

This is the module $M_{P,a,b}$ of Pakovich and Muzychuk [6, §2.1].

Number the branches at infinity so that monodromy around infinity is $(1 \ 2 \ \dots \ d)$, and put

$$\omega = e^{2\pi i/d}, \quad \mathbf{w}_j = (1, \omega^j, \omega^{2j}, \dots, \omega^{(d-1)j}).$$

For a rational subspace $M \subseteq \mathbb{Q}^d$, write

$$M_{\mathbb{C}}^{\perp} = \left\{ v \in \mathbb{C}^d : \sum_{i=1}^d m_i v_i = 0 \text{ for every } m = (m_i) \in M \right\}.$$

Thus $M_{\mathbb{C}}^{\perp}$ is taken with respect to the complex-bilinear extension of the rational dot product, not a Hermitian form.

Proposition 6.1 (Primitive Fourier inclusion; Pakovich–Muzychuk). *If $d \geq 2$, then $(M_{A,0,1})_{\mathbb{C}}$ contains $\mathbf{w}_1$ and $\mathbf{w}_{d-1}$.*

Proof. Pakovich and Muzychuk prove that $M_{A,0,1}$ contains a distinguished irreducible rational monodromy subspace W [6, Proposition 4.1 and its proof]. Their proof explicitly places $\mathbf{w}_1$ in $W_{\mathbb{C}}$. Since W is defined over $\mathbb{Q}$, it is stable under complex conjugation, so $\mathbf{w}_{d-1} = \overline{\mathbf{w}_1}$ also belongs to $W_{\mathbb{C}}$. $\square$

For $|w|$ large, $\mathcal{B}(w, t) dt$ is holomorphic in t . Deforming $[0, 1]$ to $\mu_{0,1}$ and changing variables $u = A(t)$ on each lifted edge gives

$$\mathcal{K}(w) = \sum_s \int_{\gamma_s} \sum_{i=1}^d f_{s,i} \frac{\mathcal{B}(w, t_i(u))}{A'(t_i(u))} du. \quad (6.2)$$

This is the same geometric decomposition as [6, (16)–(17)]; only the branch density differs. The edge integrals are understood as improper integrals at critical-value vertices. If u_* is such a vertex, a lifted branch tends to a point τ of ramification index e , and locally

$$t_i(u) - \tau = O((u - u_*)^{1/e}), \quad \frac{1}{A'(t_i(u))} = \frac{dt_i}{du} = O(|u - u_*|^{1/e-1}).$$

Since $1/e - 1 > -1$, these endpoint singularities are integrable. In particular, the deformation and every later edgewise integration are well-defined.

Define

$$E(t) = \exp\left(\frac{A_1(t)}{DA(t)}\right), \quad v_i(u) = \frac{E(t_i(u))}{A'(t_i(u))}. \quad (6.3)$$

For an arm γ_s , put

$$F_s(w, u) = \sum_{i=1}^d f_{s,i} \frac{\mathcal{B}(w, t_i(u))}{A'(t_i(u))}. \quad (6.4)$$

Lemma 6.2 (Terminal continuation and arm isolation). *Choose the terminal subarcs of the star pairwise disjoint except at their common regular endpoint c , and choose their interiors to avoid 0 and a prescribed finite set of points. After shortening a terminal subarc τ_s if necessary, the following hold on every continuation sheet reached from large w .*

First, in a slit neighborhood of the diagonal $w = u$ over $u \in \tau_s$, the density (6.4) has the joint expansion

$$F_s(w, u) = g_{s,0}(u) \log(w - u) + (w - u) \log(w - u) G_s(w, u) + H_s(w, u), \quad (6.5)$$

where G_s and H_s are jointly holomorphic and

$$g_{s,0}(u) = -\frac{1}{Du} \sum_{i=1}^d f_{s,i} v_i(u). \quad (6.6)$$

Second, for every interior point $w_0 \in \tau_s \setminus \{c\}$ sufficiently close to c , there is a disk D_{w_0} whose closure meets no other arm and no prescribed point. On D_{w_0} , the sum of all other arm integrals, together with the part of the s -th arm outside τ_s , is single-valued and holomorphic. Thus only the integral over τ_s can have a lateral jump across the selected arm at w_0 .

Proof. The terminal subarc contains no critical value, so the branches $t_i(u)$ and the factors $A'(t_i(u))^{-1}$ are holomorphic on a neighborhood of τ_s . The parameterized boundary symbol, including the sheet-independence proved at the end of Theorem 5.4, gives (6.5) jointly in (w, u) ; summing its lowest coefficients gives (6.6). Uniformity on the compact shortened subarc permits termwise integration.

Fix $w_0 \in \tau_s \setminus \{c\}$ sufficiently close to c . Since the terminal arms are disjoint away from c , a small closed disk around w_0 meets no other arm, no prescribed point, and no part of the selected arm outside τ_s . For every integration point on those omitted pieces one has $w \neq u$ throughout the disk. By Lemma 5.5, every already continued branch of the integrand is locally holomorphic there. The endpoint estimates preceding (6.3) give a locally uniform integrable majorant at critical-value vertices, so integration preserves holomorphy. The disk is simply connected, hence these local branches are single-valued on it. This proves the asserted isolation of the selected terminal integral. $\square$

Lemma 6.3 (Terminal lateral jump). *Let $\tau = \gamma([0, 1])$ be an oriented analytic arc, where γ extends holomorphically to a neighborhood of $[0, 1]$, $\gamma'(1) \neq 0$, and $\gamma(1) = c$. Suppose that on a slit tube around the terminal part of τ one has*

$$F(w, u) = g_0(u) \log(w - u) + (w - u) \log(w - u) G(w, u) + H(w, u),$$

with g_0 , G , and H holomorphic and with H single-valued. Let $I(w) = \int_{\tau} F(w, u) du$, and let I_+ and I_- denote its two lateral continuations across the terminal subarc. For $w = \gamma(s)$ with $s < 1$ sufficiently close to 1,

$$\begin{aligned} I_+(\gamma(s)) - I_-(\gamma(s)) &= 2\pi i \varepsilon_{\tau} \int_s^1 \left[g_0(\gamma(v)) \right. \\ &\quad \left. + (\gamma(s) - \gamma(v)) G(\gamma(s), \gamma(v)) \right] \gamma'(v) dv. \end{aligned} \tag{6.7}$$

where $\varepsilon_{\tau} \in \{\pm 1\}$ depends only on the orientation and the choice of lateral sides. In particular, if the lateral jump vanishes, then $g_0(c) = 0$.

Proof. Across the chosen slit, $\log(w - u)$ changes by the constant $2\pi i \varepsilon_{\tau}$ precisely for the portion of the oriented arc between w and the terminal endpoint c ; it has no jump on the remaining portion. This proves (6.7).

Let the integral on the right, without its nonzero constant factor, be $J(s)$. Differentiating with respect to s gives

$$J'(s) = -g_0(\gamma(s)) \gamma'(s) + R(s),$$

where $R(s) \rightarrow 0$ as $s \rightarrow 1$: every term in $R(s)$ is an integral over $[s, 1]$ of a bounded holomorphic function, and the lower-end contribution from the factor $\gamma(s) - \gamma(v)$ is zero. If the jump vanishes, then $J \equiv 0$, so

$$0 = \lim_{s \rightarrow 1} J'(s) = -g_0(c) \gamma'(1).$$

Since $\gamma'(1) \neq 0$, one obtains $g_0(c) = 0$. □

Proposition 6.4 (Rationality forces incidence orthogonality). *Assume that $\mathcal{K}$ is rational. For a sufficiently large regular center c , chosen away from 0 and the poles of $\mathcal{K}$,*

$$\sum_{i=1}^d f_{s,i} v_i(c) = 0 \quad \text{for every arm } s. \tag{6.8}$$

After analytic continuation, the same identity holds for every monodromy conjugate of every f_s . Equivalently,

$$v(c) \in (M_{A,0,1})_{\mathbb{C}}^{\perp}. \tag{6.9}$$

Proof. Choose c outside the finite critical-value set, outside 0, and outside the finite pole set of the assumed rational function $\mathcal{K}$. Choose the interiors of the terminal subarcs to avoid the latter two finite sets. On the overlap with the original large- w domain, the continued constellation sum equals the rational function $\mathcal{K}$.

Fix an arm s and a point w_0 in its terminal subarc, sufficiently close to c but distinct from c . Choose the disk D_{w_0} supplied by Lemma 6.2, small enough to contain no pole of $\mathcal{K}$. The two lateral continuations of the total constellation sum across the selected arm agree on this disk because $\mathcal{K}$ is single-valued. Every other contribution has zero jump there, so the lateral jump of

the isolated terminal integral vanishes. This holds for every such w_0 . Applying Lemma 6.3 to (6.5) gives $g_{s,0}(c) = 0$, and (6.6) yields (6.8).

It remains to justify the monodromy closure. Choose a small simply connected disk Ω of regular values around c , disjoint from 0 and from the poles of $\mathcal{K}$. The inverse branches $t_i(u)$ and the vectors $v(u)$ are holomorphic on Ω . Move the common endpoint of the star inside Ω , keeping the remote parts of the arms fixed and identifying the terminal lifts by the local covering trivialization. The incidence vectors are then locally constant. Repeating the preceding lateral-jump argument for every $c' \in \Omega$ proves

$$\sum_i f_{s,i} v_i(c') = 0 \quad (c' \in \Omega),$$

so this is an identity of analytic branches, not merely an equality at one point.

Continue that identity around loops in the regular-value locus while avoiding 0 and the finite pole set of $\mathcal{K}$. These additional punctures are regular values: every ordinary monodromy loop can be perturbed to avoid them, and a small loop around one of them has trivial covering monodromy. Thus all elements of G_A are still realized. Upon return, the inverse branches are permuted by the corresponding monodromy element σ . With the chosen left-action convention the continued relation is $(\sigma f_s) \cdot v = 0$; with the opposite convention it is $(\sigma^{-1} f_s) \cdot v = 0$, which generates the same module. Thus every monodromy conjugate is orthogonal to v , which is exactly (6.9) with the complex-bilinear convention fixed above. $\square$

If $d \geq 2$, combining Propositions 6.1 and 6.4 yields, on a sufficiently small regular-value disk Ω around c , the identity of analytic branches

$$\sum_{i=1}^d \omega^{-(i-1)} v_i(u) = 0 \quad (u \in \Omega). \quad (6.10)$$

7 The infinity obstruction and the diagonal theorem

Lemma 7.1 (Rotated exponential averages do not vanish). *Let*

$$T(U) = \alpha U + \beta + O(U^{-1}), \quad \alpha \neq 0,$$

be holomorphic for large U , let $R \in \mathbb{C}(t)$ be rational, and let $\omega = e^{2\pi i/d}$. Then

$$\sum_{j=0}^{d-1} e^{R(T(\omega^j U))} T'(\omega^j U)$$

is not identically zero near infinity.

Proof. Write $R(T(U)) = \Phi(U) + O(U^{-1})$, where Φ is the polynomial part of the Laurent expansion. Partition the indices into classes for which the polynomials $\Phi(\omega^j U)$ are identical. Every rotated phase has the same constant term $\Phi(0)$; consequently, two distinct rotated phase polynomials cannot differ by a nonzero constant. Their difference, when nonzero, has positive degree. Choose a ray avoiding the finitely many anti-Stokes directions determined by these nonzero differences. Along that ray one class C has strictly maximal real part for large U . Its contribution is

$$|C| \alpha e^{\Phi_C(U)} (1 + o(1)),$$

while every other class is exponentially smaller. This is nonzero. If all phase polynomials agree, the sum is $d \alpha e^{\Phi(U)} (1 + o(1))$, again nonzero. $\square$

Theorem 7.2 (Diagonal positive-interval constellation theorem). *Assume A_0 is nonconstant. Then*

$$\mathcal{J}(s) = \sum_{n \geq 0} \frac{I_n}{Dn + 2} s^n$$

is not rational. Consequently, $I_n \neq 0$ for infinitely many n .

Proof. Suppose $\mathcal{J}$ were rational. Then $\mathcal{K}(w) = w^{-1}\mathcal{J}(w^{-1})$ would be rational. Put $A = A_0$ and $d = \deg A$.

If $d = 1$, the lifted path from 0 to 1 is nontrivial, so some arm has $f_{s,1} \neq 0$. Proposition 6.4 would then force the nonzero density $E(t_1(c))/A'(t_1(c))$ to vanish, a contradiction.

Assume $d \geq 2$. Choose a Laurent inverse at infinity

$$A(T(U)) = U^d, \quad T(U) = \alpha U + \beta + O(U^{-1}), \quad \alpha \neq 0. \quad (7.1)$$

Choose U_0 large with $c = U_0^d$, and number the branches compatibly so that, on a connected exterior sector containing U_0 ,

$$t_i(U^d) = T(\omega^{i-1}U).$$

Put

$$R(t) = \frac{A_1(t)}{DA(t)}.$$

Differentiating (7.1) gives

$$A'(T(V))T'(V) = dV^{d-1}.$$

Hence, after pulling the analytic identity (6.10) back under $u = U^d$,

$$0 = \sum_{i=1}^d \omega^{-(i-1)} v_i(U^d) = \frac{1}{dU^{d-1}} \sum_{i=1}^d e^{R(T(\omega^{i-1}U))} T'(\omega^{i-1}U)$$

for U in a neighborhood of U_0 . The last sum is holomorphic on the connected exterior sector. By the identity theorem it vanishes identically there, contradicting Lemma 7.1. Hence $\mathcal{J}$ is not rational. If only finitely many I_n were nonzero, then $\mathcal{J}$ would be a polynomial, which is impossible. $\square$

Theorem 7.3 (Nonflat-top exclusion). *Let $0 \neq f \in \mathbb{C}[x, y]$ have exact degree D . If $A_0(t) = f_D(t, 1-t)$ is nonconstant, then $\mathcal{L}(f^n) \neq 0$ for infinitely many $n \geq 1$. In particular, f is not a counterexample to Theorem 1.1.*

Proof. Apply Theorem 7.2 to the projective layers of f . By Proposition 3.3,

$$I_n = \frac{\mathcal{L}(f^n)}{\Gamma(Dn + 2)}.$$

Thus infinitely many factorial moments are nonzero. $\square$

Corollary 7.4 (Flat-top reduction). *Every hypothetical nonzero counterexample to Theorem 1.1 has*

$$f_D(x, y) = a(x + y)^D$$

for some $a \in \mathbb{C}^\times$.

Proof. By Theorem 7.3, $f_D(t, 1-t)$ is constant. The final assertion of Lemma 3.2 gives the stated form. $\square$

8 The first flat logarithmic symbol

Assume that f has exact degree $D \geq 1$ and, after multiplication by a nonzero scalar,

$$f_D = (x + y)^D. \quad (8.1)$$

Then

$$P(t, r) = f(rt, r(1 - t)) = r^D + A_1(t)r^{D-1} + \cdots + A_D(t). \quad (8.2)$$

Define

$$F(t, u) = 1 + A_1(t)u + \cdots + A_D(t)u^D.$$

Choose a relatively compact complex neighborhood U of $[0, 1]$ and $\rho > 0$ so small that $F(t, u)$ is nonzero for $t \in U$ and $|u| < \rho$. On this product take the unique holomorphic branch

$$\Psi(t, u) = F(t, u)^{1/D}, \quad \Psi(t, 0) = 1,$$

and put

$$\phi_t(r) = r\Psi(t, r^{-1}). \quad (8.3)$$

Thus $P(t, r) = \phi_t(r)^D$.

Lemma 8.1 (Uniform flat radial inverse). *After shrinking U if necessary, there is a common exterior neighborhood on which ϕ_t has a holomorphic inverse $r = \chi(t, Y)$, jointly holomorphic in (t, Y) , satisfying*

$$\chi(t, Y) = Y - \frac{A_1(t)}{D} + O(Y^{-1}). \quad (8.4)$$

Consequently,

$$e^{Y - \chi(t, Y)} \frac{\chi(t, Y)}{Y} \partial_Y \chi(t, Y) = \exp\left(\frac{A_1(t)}{D}\right) + O(Y^{-1}), \quad (8.5)$$

locally uniformly in $t \in U$.

Proof. Locally uniformly for $t \in U$,

$$\phi_t(r) = r + \frac{A_1(t)}{D} + O(r^{-1}).$$

In the coordinate $z = r^{-1}$, the map $z \mapsto 1/\phi_t(1/z)$ has derivative 1 at $z = 0$. The parameter-dependent inverse-function theorem therefore gives a common inverse at infinity and (8.4). Substitution into the left side of (8.5) gives the displayed constant term. $\square$

Lemma 8.2 (Common-ray comparison). *After shrinking U and increasing R , there are a common exterior sector $\mathfrak{S}$ and a ray $\ell = Y_0 + [0, \infty) \subset \mathfrak{S}$ such that all curves $\Gamma_t = \phi_t([R, \infty))$, $t \in U$, lie in $\mathfrak{S}$ and have bounded displacement from ℓ . Let $F(t, Y)$ be jointly holomorphic on $U \times \mathfrak{S}$, analytic in Y^{-1} at infinity, and satisfy*

$$F(t, Y) = q_0(t) + O(Y^{-1})$$

locally uniformly in t . For η in a sufficiently small decay sector, the difference

$$\int_{\Gamma_t} e^{-\eta Y} F(t, Y) dY - \int_{\ell} e^{-\eta Y} F(t, Y) dY$$

extends jointly holomorphically across $\eta = 0$.

Proof. The expansion of ϕ_t shows that the curves, the common ray, and the initial and outer connectors used below all lie in one common exterior sector. Write $F = q_0 + \tilde{F}$ with $\tilde{F} = O(Y^{-1})$. For the constant term, path independence of a primitive of $e^{-\eta Y}$ gives

$$q_0(t) \frac{e^{-\eta a_t} - e^{-\eta Y_0}}{\eta}, \quad a_t = \phi_t(R),$$

which is holomorphic at $\eta = 0$. For $\tilde{F}$, truncate the two curves at a common large ray parameter. Their outer endpoints remain at bounded distance, so the outer connector has bounded length and contributes $O(|Y|^{-1})$, uniformly on closed decay subsectors; this tends to zero. Cauchy's theorem, now applied inside the domain of holomorphy $U \times \mathfrak{S}$, reduces the difference to a finite initial-connector integral, which is jointly holomorphic. $\square$

Define

$$\mathcal{G}_f(s) = \sum_{n \geq 0} \frac{\mathcal{L}(f^n)}{\Gamma(Dn + 2)} s^n. \quad (8.6)$$

Proposition 8.3 (First flat singular coefficient). *Let $\lambda^D = s$ near $\lambda = 1$ and put $\eta = 1 - \lambda$. On a fixed admissible sector at $\eta = 0$,*

$$\mathcal{G}_f(\lambda^D) = \frac{c_0(f)}{D\lambda\eta} + O(\log \eta), \quad (8.7)$$

where the error is the sum of a holomorphic germ and a function bounded by $C(1 + |\log \eta|)$ on every smaller closed sector, and

$$c_0(f) = \int_0^1 \exp\left(\frac{A_1(t)}{D}\right) dt. \quad (8.8)$$

In particular,

$$\mathcal{L}(f^n) = 0 \quad (n \geq 1) \quad \implies \quad c_0(f) = 0. \quad (8.9)$$

Proof. By Proposition 4.3,

$$\mathcal{G}_f(s) = \int_0^1 \int_0^\infty e^{-r} r E_{D,2}(sP(t, r)) dr dt.$$

Split the radial integral at a sufficiently large R and apply Lemma 4.4 with $m = 1$ on the tail, where $\phi_t(r)$ is uniformly nonzero. The compact radial part is holomorphic near $\lambda = 1$. The low-degree correction vanishes for $D \geq 2$; when $D = 1$, $C_{1,1}(X) = X^{-1}$ and its tail contribution is

$$-\frac{1}{\lambda} \int_0^1 \int_R^\infty \frac{e^{-r}}{\Psi(t, r^{-1})} dr dt,$$

which is also holomorphic near $\lambda = 1$. Every root-filter term with $k \neq 0$ is holomorphic there because $\Re(1 - \omega^k \lambda) > 0$. The only singular term is

$$\frac{1}{D\lambda} \int_0^1 \int_R^\infty \frac{e^{-r + \lambda \phi_t(r)}}{\Psi(t, r^{-1})} dr dt. \quad (8.10)$$

Set $Y = \phi_t(r)$ and $r = \chi(t, Y)$. Since $\Psi(t, r^{-1}) = Y/r$, (8.10) becomes

$$\frac{1}{D\lambda} \int_0^1 \int_{\Gamma_t} e^{-\eta Y} e^{Y - \chi(t, Y)} \frac{\chi(t, Y)}{Y} \partial_Y \chi(t, Y) dY dt.$$

By Lemmas 8.1 and 8.2, modulo a function holomorphic at $\eta = 0$ this equals

$$\frac{1}{D\lambda} \int_{Y_0}^{\infty} e^{-\eta Y} \left[c_0(f) + O(Y^{-1}) \right] dY.$$

The constant term contributes $c_0(f)e^{-\eta Y_0}/(D\lambda\eta)$, whose difference from the displayed pole in (8.7) is holomorphic. The Laplace transform of the $O(Y^{-1})$ remainder is $O(1 + |\log \eta|)$ on smaller closed sectors. This proves (8.7) and (8.8).

For real $0 < \lambda < 1$, the root-filter integrals above converge absolutely and agree algebraically with the original Mittag–Leffler integral. Hence the sectorial germ just computed is the analytic continuation, along the positive interval, of the original germ of $\mathcal{G}_f(\lambda^D)$ at $\lambda = 0$.

If all positive factorial moments vanish, then that original germ is $\mathcal{G}_f(s) \equiv 1$, and uniqueness of analytic continuation makes the continued germ constant as well. A constant function has no pole at $\lambda = 1$, so (8.7) forces $c_0(f) = 0$. $\square$

9 Algebraic exponential integrals

Theorem 9.1 (Algebraic exponential-integral theorem). *Let $q \in \overline{\mathbb{Q}}[x]$ be nonconstant, and let $A, B \in \overline{\mathbb{Q}}$ with $A \neq B$. Then*

$$\int_A^B e^{q(x)} dx$$

is transcendental. In particular, it is nonzero.

The integral is independent of the path because the integrand is entire. The rest of the paper proves this theorem before returning to Theorem 1.1.

10 E -functions and algebraic normalization

We use the following definition, in the form adopted by Beukers [2, §1].

Definition 10.1 (E -function). An entire function

$$F(z) = \sum_{m \geq 0} a_m \frac{z^m}{m!}$$

is an E -function if:

- (i) $a_m \in \overline{\mathbb{Q}}$ for every m ;
- (ii) the logarithmic height $h(a_0, \dots, a_m)$ is $O(m)$;
- (iii) F satisfies a nonzero linear differential equation with coefficients in $\overline{\mathbb{Q}}[z]$.

The key arithmetic input is Beukers' lifting theorem.

Theorem 10.2 (Beukers). *Let $F_1, \dots, F_N$ be E -functions forming a vector solution of*

$$\mathbf{F}' = A(z)\mathbf{F}, \quad A(z) \in M_N(\overline{\mathbb{Q}}(z)),$$

and let $T(z) \in \overline{\mathbb{Q}}[z]$ be a common denominator of the entries of $A(z)$. Let $\xi \in \overline{\mathbb{Q}}$ satisfy $\xi T(\xi) \neq 0$. If $P \in \overline{\mathbb{Q}}[X_1, \dots, X_N]$ is homogeneous and

$$P(F_1(\xi), \dots, F_N(\xi)) = 0,$$

then there is a polynomial $Q \in \overline{\mathbb{Q}}[z, X_1, \dots, X_N]$, homogeneous in the variables $X_1, \dots, X_N$ and of the same degree as P , such that

$$Q(z, F_1(z), \dots, F_N(z)) \equiv 0, \quad Q(\xi, X_1, \dots, X_N) = P(X_1, \dots, X_N).$$

In particular, a linear relation at $z = \xi$ lifts to a linear functional relation whose coefficient polynomials specialize coefficientwise to the prescribed relation. No hypothesis of $\mathbb{C}(z)$ -linear independence of the F_i is required.

This is [2, Theorem 1.3]; its linear-relation form is [2, Theorem 3.2].

We shall also use the elementary independence of distinct exponentials.

Lemma 10.3 (Distinct exponentials). *If $\lambda_1, \dots, \lambda_s \in \mathbb{C}$ are distinct, then*

$$e^{\lambda_1 z}, \dots, e^{\lambda_s z}$$

are linearly independent over $\mathbb{C}(z)$.

Proof. Suppose a nontrivial relation with the fewest nonzero terms exists. Differentiate it and eliminate one term. Minimality then forces the logarithmic derivative of a nonzero rational function to be a nonzero constant. This is impossible, since a logarithmic derivative has no nonzero constant term at infinity. $\square$

10.1 Algebraic normalization

Let q_0 have degree $n \geq 2$. Translation by an algebraic number removes the coefficient of x^{n-1} , and an algebraic nonzero scaling of the variable normalizes the leading coefficient to $1/n$. Thus there are $\alpha \in \overline{\mathbb{Q}}^\times$ and $\beta \in \overline{\mathbb{Q}}$ such that

$$q(y) := q_0(\alpha y + \beta) = \frac{y^n}{n} + a_{n-2}y^{n-2} + \dots + a_1y + a_0, \quad a_j \in \overline{\mathbb{Q}}. \quad (10.1)$$

The change of variable $x = \alpha y + \beta$ gives

$$\int_A^B e^{q_0(x)} dx = \alpha \int_{(A-\beta)/\alpha}^{(B-\beta)/\alpha} e^{q(y)} dy.$$

Hence the transformed endpoints remain distinct and algebraic, and the original integral differs from the normalized one by a nonzero algebraic factor. Since such multiplication preserves transcendence, it suffices to prove Theorem 9.1 under (10.1). We impose this normalization through Section 17.

11 The intrinsic moment connection

Assume (10.1), put $n = \deg q$, and set

$$K = \overline{\mathbb{Q}}(z).$$

Define the twisted de Rham quotient

$$\mathcal{M}_q = K[x]/(\partial_x + zq'(x))K[x]. \quad (11.1)$$

We write $[p]$ for the class of $p \in K[x]$.

Lemma 11.1 (Brieskorn basis). *The classes*

$$e_j = [x^j], \quad 0 \leq j \leq n-2,$$

form a K -basis of $\mathcal{M}_q$. More precisely,

$$\mathcal{G}_q = \mathbb{C}[z, z^{-1}, x] / (\partial_x + zq'(x))\mathbb{C}[z, z^{-1}, x]$$

is a free $\mathbb{C}[z, z^{-1}]$ -module with the same basis. Differentiation in z defines the connection

$$\nabla_{\partial_z}[p] = [\partial_z p + qp]. \quad (11.2)$$

Proof. The relation

$$[zq'(x)p(x)] = -[p'(x)]$$

reduces every monomial of degree at least $n-1$ to lower degree, both over K and over $\mathbb{C}[z, z^{-1}]$. Thus the displayed classes span.

If $0 \neq p \in K[x]$, then $(\partial_x + zq')p$ has x -degree $\deg_x p + n - 1$, and its leading term cannot cancel. The same argument applies over $\mathbb{C}[z, z^{-1}]$. Hence the image contains no nonzero polynomial of degree at most $n-2$, proving independence.

The connection is well defined because

$$[\partial_z + q, \partial_x + zq'] = 0 :$$

indeed, the commutators $[\partial_z, zq'] = q'$ and $[q, \partial_x] = -q'$ cancel. Formula (11.2) is the algebraic form of differentiating the kernel e^{zq} . $\square$

For $0 \leq j \leq n-2$, perform Euclidean division

$$q(x)x^j = h_j(x)q'(x) + r_j(x), \quad \deg r_j \leq n-2. \quad (11.3)$$

Let $\mathbf{C}$ and $\mathbf{D}$ be the matrices whose j th columns are the coefficient vectors of r_j and $-h'_j$ in the basis $1, x, \dots, x^{n-2}$.

Lemma 11.2 (The matrices $\mathbf{C}$ and $\mathbf{D}$). *The following assertions hold.*

(i) $\mathbf{C}$ is multiplication by q on the Jacobian algebra $\overline{\mathbb{Q}}[x]/(q')$;

(ii) $\mathbf{D}$ is triangular with diagonal

$$-\frac{1}{n}, -\frac{2}{n}, \dots, -\frac{n-1}{n};$$

(iii) the connection matrix on $\mathcal{M}_q$ is $\mathbf{C} + \mathbf{D}/z$.

Proof. In $\mathcal{M}_q$,

$$[h_j q'] = -\frac{1}{z}[h'_j],$$

so (11.3) gives

$$[qx^j] = [r_j] - \frac{1}{z}[h'_j].$$

This proves (i) and (iii).

Because q has no x^{n-1} term and q' has no x^{n-2} term, division gives

$$h_j(x) = \frac{x^{j+1}}{n} + \text{a polynomial of degree at most } j-1.$$

Therefore

$$-h'_j(x) = -\frac{j+1}{n}x^j + \text{terms of degree at most } j-2,$$

which proves (ii). $\square$

The degree of q is denoted by n throughout the exponential-integral argument. In Sections 11 to 14, the capital letter D introduced below denotes a residue matrix and is unrelated to the bivariate degree used in Sections 3 and 8.

Let $A \neq B$ be algebraic, and define

$$U_j(z) = \int_A^B x^j e^{zq(x)} dx, \quad \mathbf{U} = (U_0, \dots, U_{n-2})^\top. \quad (11.4)$$

Define the endpoint vector

$$\mathbf{b}(x) = n(h_0(x), \dots, h_{n-2}(x))^\top. \quad (11.5)$$

Since $h_0(x) = x/n$, the first coordinate of $\mathbf{b}(x)$ is x .

Let

$$\Lambda_\partial = \{q(A), q(B)\}$$

with repetitions removed, and put

$$\mathbf{b}_\lambda = \sum_{\substack{x \in \{A, B\} \\ q(x) = \lambda}} \varepsilon_x \mathbf{b}(x), \quad \varepsilon_B = 1, \quad \varepsilon_A = -1. \quad (11.6)$$

Proposition 11.3 (Moment system). *Put $C = C^\top$ and $D = D^\top$. Then*

$$\mathbf{U}' = \left(C + \frac{D}{z}\right) \mathbf{U} + \frac{1}{nz} \sum_{\lambda \in \Lambda_\partial} \mathbf{b}_\lambda e^{\lambda z}. \quad (11.7)$$

Moreover,

$$\text{Spec}(D) = \left\{-\frac{1}{n}, -\frac{2}{n}, \dots, -\frac{n-1}{n}\right\}, \quad (11.8)$$

and

$$\sum_{\lambda \in \Lambda_\partial} (\mathbf{b}_\lambda)_0 = B - A \neq 0. \quad (11.9)$$

Proof. From (11.3),

$$\begin{aligned} U'_j &= \int_A^B q(x) x^j e^{zq(x)} dx \\ &= \int_A^B \left(r_j(x) - \frac{1}{z} h'_j(x) \right) e^{zq(x)} dx + \frac{1}{z} \left[h_j(x) e^{zq(x)} \right]_A^B. \end{aligned}$$

Writing these identities simultaneously gives (11.7). The spectral assertion follows from Lemma 11.2. The first coordinate of $\mathbf{b}(x)$ is x , which gives (11.9). $\square$

Lemma 11.4 (Joint height from one common denominator). *Let K be a number field, let $\alpha_0, \dots, \alpha_m \in K$, and let $\delta \in \mathbb{Z}_{\geq 1}$ satisfy $\delta \alpha_k \in \mathcal{O}_K$ for every k . Then*

$$h(\alpha_0, \dots, \alpha_m) \leq \log \delta + \log \max \left(1, \max_{0 \leq k \leq m} \max_{\sigma: K \hookrightarrow \mathbb{C}} |\sigma(\alpha_k)| \right).$$

Here the left side is the absolute logarithmic Weil height of $(1 : \alpha_0 : \dots : \alpha_m)$.

Proof. With the standard absolute values and local degrees $n_v = [K_v : \mathbb{Q}_v]$,

$$h(\alpha_0, \dots, \alpha_m) = \frac{1}{[K : \mathbb{Q}]} \sum_{v \in M_K} n_v \log \max(1, |\alpha_0|_v, \dots, |\alpha_m|_v).$$

Put $\beta_k = \delta \alpha_k \in \mathcal{O}_K$. Projective invariance gives

$$(1 : \alpha_0 : \dots : \alpha_m) = (\delta : \beta_0 : \dots : \beta_m).$$

At every finite place, all displayed coordinates have absolute value at most 1, so the corresponding logarithm is at most zero. At an infinite place,

$$\max(|\delta|_v, |\beta_0|_v, \dots, |\beta_m|_v) \leq \delta \max \left(1, \max_{k, \sigma} |\sigma(\alpha_k)| \right).$$

Since $\sum_{v|\infty} n_v = [K : \mathbb{Q}]$, summing these upper bounds proves the claim. $\square$

Lemma 11.5 (Moment functions are E -functions). *For $0 \leq j \leq n-2$, the entire function U_j in (11.4) is an E -function. More precisely, if*

$$U_j(z) = \sum_{m \geq 0} u_{j,m} \frac{z^m}{m!}, \quad u_{j,m} = \int_A^B x^j q(x)^m dx,$$

then there is a constant κ_j such that

$$h(u_{j,0}, \dots, u_{j,m}) \leq \kappa_j(m+1) \quad (m \geq 0),$$

and U_j satisfies a nonzero homogeneous linear differential equation over $\overline{\mathbb{Q}}[z]$ of order at most $n+1$.

Proof. We integrate over the straight segment from A to B ; the value is path independent because the integrand is entire in x . Uniform convergence on compact subsets gives

$$U_j(z) = \sum_{m \geq 0} u_{j,m} \frac{z^m}{m!}, \quad u_{j,m} = \int_A^B x^j q(x)^m dx,$$

and also proves that U_j is entire.

Let

$$K_0 = \mathbb{Q}(A, B, \text{the coefficients of } q).$$

Write

$$q(x) = \sum_{i=0}^n c_i x^i, \quad q(x)^m = \sum_{k=0}^{nm} \gamma_{m,k} x^k.$$

For $m \geq 1$,

$$\gamma_{m,k} = \sum_{i_1 + \dots + i_m = k} c_{i_1} \cdots c_{i_m},$$

while $\gamma_{0,0} = 1$. Termwise integration gives the exact formula

$$u_{j,m} = \sum_{k=0}^{nm} \gamma_{m,k} \frac{B^{j+k+1} - A^{j+k+1}}{j+k+1}. \quad (11.10)$$

It follows immediately that $u_{j,m} \in K_0$.

We next bound all archimedean conjugates simultaneously. Put

$$M = \max_{\sigma: K_0 \hookrightarrow \mathbb{C}} \max(1, |\sigma(A)|, |\sigma(B)|),$$

$$\Theta = \max_{\sigma: K_0 \hookrightarrow \mathbb{C}} \max_{|x| \leq M} |q^\sigma(x)|, \quad \Delta = \max_{\sigma: K_0 \hookrightarrow \mathbb{C}} |\sigma(B) - \sigma(A)|.$$

Every conjugate segment lies in $|x| \leq M$, and applying σ to (11.10) gives the corresponding conjugate integral. Hence

$$|\sigma(u_{j,m})| \leq \Delta M^j \Theta^m.$$

With $C_1 = \max(1, \Delta, M, \Theta)$, this is at most C_1^{m+j+1} for every embedding σ .

Choose $d \in \mathbb{Z}_{\geq 1}$ so that

$$dc_0, \dots, dc_n, dA, dB \in \mathcal{O}_{K_0}.$$

For $m \geq 0$, set

$$\delta_m = d^{(n+1)m+j+1} \text{lcm}(1, \dots, nm+j+1).$$

Then $d^m \gamma_{m,k}$ and $d^{j+k+1}(B^{j+k+1} - A^{j+k+1})$ are algebraic integers. Since $0 \leq k \leq nm$, the k th summand of (11.10), after multiplication by δ_m , is

$$(d^m \gamma_{m,k}) d^{nm-k} \left(d^{j+k+1} (B^{j+k+1} - A^{j+k+1}) \right) \frac{\text{lcm}(1, \dots, nm+j+1)}{j+k+1},$$

a product of algebraic integers. Thus $\delta_m u_{j,m}$ is integral. Moreover $\delta_\ell \mid \delta_m$ whenever $0 \leq \ell \leq m$, so δ_m is a common denominator for $u_{j,0}, \dots, u_{j,m}$.

By Lemma 11.4,

$$h(u_{j,0}, \dots, u_{j,m}) \leq \log \delta_m + (m+j+1) \log C_1.$$

The elementary Chebyshev bound $\log \text{lcm}(1, \dots, N) \leq 2N$ gives

$$\log \delta_m \leq ((n+1)m+j+1) \log d + 2(nm+j+1).$$

Since j is fixed, these inequalities give $h(u_{j,0}, \dots, u_{j,m}) = O(m)$.

It remains to establish a scalar differential equation. Order the distinct endpoint values as $\lambda_1, \dots, \lambda_s$, where $s \leq 2$, and let $\mathcal{W}$ be the $\overline{\mathbb{Q}}(z)$ -span of

$$U_0, \dots, U_{n-2}, e^{\lambda_1 z}, \dots, e^{\lambda_s z}$$

inside the field of meromorphic functions on $\mathbb{C} \setminus \{0\}$. By Proposition 11.3, this space is stable under ∂_z , and

$$\dim_{\overline{\mathbb{Q}}(z)} \mathcal{W} \leq n - 1 + s \leq n + 1.$$

Therefore the $n + 2$ functions $U_j, \partial_z U_j, \dots, \partial_z^{n+1} U_j$ are linearly dependent over $\overline{\mathbb{Q}}(z)$. Clearing denominators gives a nonzero operator in $\overline{\mathbb{Q}}[z]\langle \partial_z \rangle$ of order at most $n + 1$ annihilating U_j . The three defining conditions of an E -function now follow. $\square$

12 Semisimplicity of the moment connection

Let Σ be the finite set of branch values of q , and set

$$V = \mathbb{A}_t^1 \setminus \Sigma.$$

Over V , the map $q : q^{-1}(V) \rightarrow V$ is a degree- n unramified covering. Let $\mathcal{L}$ be its permutation local system and

$$\mathcal{L}^0 = \ker(\text{Tr} : \mathcal{L} \rightarrow \mathbb{C}_V)$$

its reduced, or augmentation, local system of rank $n - 1$.

Proposition 12.1 (Semisimple Fourier–Laplace connection). *After extension of constants to $\mathbb{C}$, the differential module over $\mathbb{C}(z)$ defined by the homogeneous system*

$$\mathbf{Y}' = \left(C + \frac{D}{z} \right) \mathbf{Y} \tag{12.1}$$

is semisimple.

Proof. The cover $q^{-1}(V) \rightarrow V$ is connected, because $q^{-1}(V)$ is obtained from the affine line by deleting finitely many points. Its monodromy action on a fiber is therefore transitive. The permutation local system has the trace splitting

$$\mathcal{L} \simeq \mathbb{C}_V \oplus \mathcal{L}^0 :$$

indeed, the diagonal map followed by the trace is multiplication by n , which is invertible over $\mathbb{C}$. The monodromy image is finite, so Maschke’s theorem makes $\mathcal{L}^0$ semisimple. Write

$$\mathcal{L}^0 = \bigoplus_{\alpha} \mathcal{L}_{\alpha}$$

as a direct sum of irreducible local systems.

A finite morphism has no positive-dimensional fibers, so the defining condition for smallness is vacuous; hence q is small. The proper small-map theorem gives

$$Rq_* \mathbb{C}_{\mathbb{A}_x^1}[1] \simeq \text{IC}(\mathcal{L});$$

see [3, Proposition 4.2.1 and Remark 4.2.4]. The intermediate extensions $\text{IC}(\mathcal{L}_{\alpha})$ are simple perverse sheaves, while $\text{IC}(\mathbb{C}_V) = \mathbb{C}_{\mathbb{A}_t^1}[1]$. Under the Riemann–Hilbert correspondence they

correspond to simple regular holonomic $\mathcal{D}_{\mathbb{A}_t^1}$ -modules; see [4, §§3.4, 7.1–7.2, and 8.2.2]. Since the direct image above is perverse, the derived D -module direct image is concentrated in degree zero, and compatibility with proper direct image yields

$$q_+ \mathcal{O}_{\mathbb{A}_x^1} \simeq \mathcal{O}_{\mathbb{A}_t^1} \oplus \mathcal{N}^0, \quad (12.2)$$

where $\mathcal{N}^0$ is a direct sum of simple regular holonomic modules.

Use a Fourier variable τ and the Weyl-algebra convention

$$t \longmapsto -\partial_\tau, \quad \partial_t \longmapsto \tau. \quad (12.3)$$

It corresponds to the kernel $e^{-\tau t}$. This is an isomorphism of Weyl algebras, so Fourier transform is an exact equivalence and preserves simple objects and direct sums. The transform of the constant connection is

$$\widehat{\mathcal{O}_{\mathbb{A}_t^1}} \simeq \mathcal{D}_{\mathbb{A}_\tau^1} / \mathcal{D}_{\mathbb{A}_\tau^1} \tau,$$

which is supported at $\tau = 0$ and disappears after localization to $\mathbb{C}(\tau)$.

We record why generic localization preserves each surviving simple summand. Let M be a simple Weyl module and let $S = \mathbb{C}[\tau] \setminus \{0\}$. Its S -torsion is a Weyl submodule: if $p(\tau)m = 0$, then $p(\tau)^2 \partial_\tau m = 0$. Thus either $S^{-1}M = 0$, or $M \hookrightarrow S^{-1}M$. In the latter case, a nonzero differential submodule $N \subseteq S^{-1}M$ meets M nontrivially after clearing a denominator. The intersection $N \cap M$ is a nonzero Weyl submodule, hence is M , and therefore $N = S^{-1}M$. Thus every nonzero localized simple summand is irreducible.

It remains to identify the localized Fourier transform and its signs. Factor q as its graph embedding followed by the projection to the t -line. The graph module is generated by a symbol δ_q satisfying

$$(t - q(x))\delta_q = 0, \quad (\partial_x + q'(x)\partial_t)\delta_q = 0.$$

After (12.3), the transformed generator satisfies

$$(\partial_\tau + q(x))\widehat{\delta}_q = 0, \quad (\partial_x + \tau q'(x))\widehat{\delta}_q = 0;$$

it is the exponential generator $e^{-\tau q(x)}$. On its coefficient polynomials, the relative x -differential and the τ -connection are therefore

$$\partial_x - \tau q'(x), \quad \partial_\tau - q(x),$$

respectively. The projection along the x -line is computed by the relative de Rham complex

$$\left[\mathbb{C}(\tau)[x] \xrightarrow{\partial_x - \tau q'} \mathbb{C}(\tau)[x] \right],$$

with the left term in degree -1 and the right term in degree 0 . Its negative-degree cohomology is the kernel, which is zero: for nonzero $p \in \mathbb{C}(\tau)[x]$, the term $-\tau q'p$ has strictly larger x -degree than p . Consequently the degree-zero generic Fourier transform is the cokernel

$$\mathbb{C}(\tau)[x] / (\partial_x - \tau q'(x))\mathbb{C}(\tau)[x], \quad \nabla_{\partial_\tau}[p] = [\partial_\tau p - qp]. \quad (12.4)$$

Pulling back by $\tau = -z$ changes the relative operator to $\partial_x + zq'$ and, because $\partial_z = -\partial_\tau$, changes the connection to $\partial_z + q$. Thus (12.4) becomes $\mathcal{M}_q$ with the connection (11.2). By Lemma 11.2,

its matrix in the monomial basis is $\mathbf{C} + \mathbf{D}/z$. It follows from the preceding decomposition and localization argument that $\mathcal{M}_q \otimes_{\overline{\mathbb{Q}}(z)} \mathbb{C}(z)$ is semisimple.

For clarity, if a connection has matrix M in a basis (e_j) , so that $\nabla e_j = \sum_i M_{ij} e_i$, a horizontal functional has value column $\mathbf{y} = (\phi(e_j))_j$ satisfying $\mathbf{y}' = M^\top \mathbf{y}$. Thus the homogeneous period system has coefficient matrix

$$\mathbf{C}^\top + \frac{\mathbf{D}^\top}{z} = C + \frac{D}{z}.$$

The row module with connection $\phi \mapsto \phi' + \phi(C + D/z)$ used below is its dual. Duality preserves semisimplicity, proving the proposition. $\square$

13 Horizontal-endomorphism rigidity

We now prove the main structural input. The proof is entirely algebraic and treats degenerate critical points and coincident critical values uniformly.

Let

$$\mathcal{V} = \mathbb{C}[x]_{<n-1}.$$

For $g \in \mathcal{V}$, divide

$$gq = h_g q' + r_g, \quad \deg r_g < n-1, \quad (13.1)$$

and define linear operators

$$\mathbf{C}_* g = r_g, \quad \mathbf{D}_* g = -h'_g. \quad (13.2)$$

The quotient and remainder depend linearly on g by uniqueness of Euclidean division. Moreover,

$$\deg h_g \leq \deg g + 1 \leq n-1, \quad \deg(-h'_g) \leq n-2,$$

so both operators map $\mathcal{V}$ to itself. They are represented by $\mathbf{C}$ and $\mathbf{D}$ in the monomial basis.

For a critical point τ , put

$$m_\tau = \text{ord}_\tau q', \quad \lambda_\tau = q(\tau).$$

Lemma 13.1 (Critical-point basis and local spectrum). *For every critical point τ and $1 \leq k \leq m_\tau$, define*

$$g_{\tau,k}(x) = \frac{q'(x)}{(x-\tau)^k}. \quad (13.3)$$

Then:

(i) *the polynomials $g_{\tau,k}$, over all τ and k , form a basis of $\mathcal{V}$;*

(ii) *$\mathbf{C}_*$ is semisimple and*

$$\mathbf{C}_* g_{\tau,k} = \lambda_\tau g_{\tau,k};$$

(iii) *put*

$$E_\tau = \text{span}_{\mathbb{C}}\{g_{\tau,1}, \dots, g_{\tau,m_\tau}\}.$$

If $E_\lambda = \ker(\mathbf{C}_ - \lambda I)$ and π_λ is the spectral projection, then*

$$E_\lambda = \bigoplus_{\tau: \lambda_\tau = \lambda} E_\tau, \quad \overline{\mathbf{D}}_\lambda = \pi_\lambda \mathbf{D}_*|_{E_\lambda}$$

preserves every summand E_τ . In the ordered basis $g_{\tau,1}, \dots, g_{\tau,m_\tau}$, its restriction to E_τ is triangular with diagonal

$$-\frac{m_\tau}{m_\tau+1}, -\frac{m_\tau-1}{m_\tau+1}, \dots, -\frac{1}{m_\tau+1}. \quad (13.4)$$

Proof. Factor

$$q'(x) = c \prod_{\tau} (x - \tau)^{m_{\tau}}.$$

Reduction modulo q' identifies $\mathcal{V}$ with the Jacobian algebra, and the Chinese remainder theorem gives

$$\mathbb{C}[x]/(q') \simeq \bigoplus_{\tau} \mathbb{C}[x]/(x - \tau)^{m_{\tau}}.$$

The E_{σ} -component of an element of $\mathcal{V}$ is precisely its residue class modulo $(x - \sigma)^{m_{\sigma}}$. For fixed τ , the images of $g_{\tau,k}$ vanish in every other local factor, while in the τ -factor, with $y = x - \tau$, they are

$$g_{\tau,k} = u_{\tau}(y)y^{m_{\tau}-k}, \quad u_{\tau}(0) \neq 0.$$

Hence $g_{\tau,1}, \dots, g_{\tau,m_{\tau}}$ form a basis of the τ -factor. Since $\sum_{\tau} m_{\tau} = n - 1 = \dim \mathcal{V}$, all the displayed vectors form a basis of $\mathcal{V}$, proving (i).

Since $q - \lambda_{\tau}$ vanishes at τ to order $m_{\tau} + 1$, the polynomial

$$h_{\tau,k}(x) = \frac{q(x) - \lambda_{\tau}}{(x - \tau)^k}$$

is well defined and satisfies

$$qg_{\tau,k} = h_{\tau,k}q' + \lambda_{\tau}g_{\tau,k}.$$

Thus $\mathbb{C}_*g_{\tau,k} = \lambda_{\tau}g_{\tau,k}$. The basis just constructed is therefore an eigenbasis for $\mathbb{C}_*$, proving (ii), and

$$E_{\lambda} = \bigoplus_{\tau: \lambda_{\tau} = \lambda} E_{\tau}.$$

Write $m = m_{\tau}$, $y = x - \tau$, and

$$q(x) - \lambda_{\tau} = ay^{m+1} + O(y^{m+2}), \quad a \neq 0.$$

Then

$$g_{\tau,k} = a(m+1)y^{m-k} + O(y^{m-k+1})$$

and

$$-h'_{\tau,k} = -a(m+1-k)y^{m-k} + O(y^{m-k+1}).$$

In the ordered local basis, the error term can involve only $g_{\tau,j}$ with $j < k$: it has no local term of order below y^{m-k} . Consequently the compressed local block is triangular and its k th diagonal entry is

$$-\frac{m+1-k}{m+1}.$$

It remains to exclude cross terms between distinct critical points over the same critical value. If $\sigma \neq \tau$ and $\lambda_{\sigma} = \lambda_{\tau}$, then

$$h_{\tau,k} = \frac{q - \lambda_{\tau}}{(x - \tau)^k}$$

vanishes at σ to order $m_{\sigma} + 1$, because the denominator is a unit there. Hence $h'_{\tau,k}$ vanishes modulo $(x - \sigma)^{m_{\sigma}}$. Thus the compression $\pi_{\lambda} \mathbb{D}_*|_{E_{\lambda}}$ preserves each E_{τ} and has the claimed diagonal spectrum. $\square$

Remark 13.2. There is no conflict between (11.8) and (13.4). The first is the spectrum of the full operator D_* , whereas the second describes its block-diagonal compression relative to the spectral decomposition of C_* . Their traces agree:

$$-\sum_{\tau} \sum_{r=1}^{m_{\tau}} \frac{r}{m_{\tau}+1} = -\frac{1}{2} \sum_{\tau} m_{\tau} = -\frac{n-1}{2} = -\sum_{j=1}^{n-1} \frac{j}{n}.$$

Theorem 13.3 (Horizontal-endomorphism rigidity). *Let $P(z) \in M_{n-1}(\mathbb{C}(z))$ satisfy*

$$P' = \left[C + \frac{D}{z}, P \right]. \quad (13.5)$$

Then P is constant. Consequently, the horizontal endomorphism matrices of the homogeneous moment system (12.1) are exactly

$$Z(C) \cap Z(D),$$

and all are constant. Equivalently, in the monomial basis the horizontal endomorphism matrices of the algebraic connection $\mathcal{M}_q$ are exactly

$$Z(C) \cap Z(D).$$

The same constancy conclusion holds for the corresponding dual modules.

Proof. First exclude finite poles. If P had a pole of order $m \geq 1$ at $z_0 \neq 0$, then P' would have pole order $m+1$, whereas the right side of (13.5) would have order at most m . Thus no such pole occurs.

At $z = 0$, suppose

$$P = z^{-m} P_{-m} + O(z^{-m+1}), \quad m \geq 1, \quad P_{-m} \neq 0.$$

The coefficient of z^{-m-1} is

$$-m P_{-m} = [D, P_{-m}].$$

The eigenvalues of D are $-j/n$, $1 \leq j \leq n-1$, so the spectrum of $\text{ad } D : X \mapsto [D, X]$ consists of their pairwise differences and is contained strictly in $(-1, 1)$. It does not contain the negative integer $-m$, a contradiction. Hence P has no finite pole. Since P is rational, it is a polynomial matrix.

We now exclude positive degree. Transpose (13.5) and put $Q = P^T$. Then

$$Q' = [Q, C_* + D_*/z]. \quad (13.6)$$

Suppose

$$Q(z) = Q_d z^d + Q_{d-1} z^{d-1} + \cdots, \quad d \geq 1, \quad Q_d \neq 0.$$

Multiplying (13.6) by z and comparing the coefficients of z^{d+1} and z^d gives

$$[Q_d, C_*] = 0, \quad (13.7)$$

$$dQ_d = [Q_{d-1}, C_*] + [Q_d, D_*]. \quad (13.8)$$

This part of the argument uses only the matrix semisimplicity of C_* proved in Lemma 13.1, not the differential-module semisimplicity of Proposition 12.1.

For each critical value λ , its spectral projection is a polynomial in C_* :

$$\pi_\lambda = \prod_{\mu \neq \lambda} \frac{C_* - \mu I}{\lambda - \mu}.$$

Equation (13.7) therefore implies that Q_d commutes with every π_λ and is block diagonal. Put

$$Q_{d,\lambda} = \pi_\lambda Q_d \pi_\lambda.$$

For at least one λ , this block is nonzero. Applying π_λ on both sides of (13.8) kills the $[Q_{d-1}, C_*]$ term and gives

$$\pi_\lambda [Q_d, D_*] \pi_\lambda = [Q_{d,\lambda}, \pi_\lambda D_* \pi_\lambda] = [Q_{d,\lambda}, \overline{D}_\lambda].$$

Thus

$$dQ_{d,\lambda} = [Q_{d,\lambda}, \overline{D}_\lambda]. \quad (13.9)$$

After triangularizing $\overline{D}_\lambda$, the commutator operator $X \mapsto [X, \overline{D}_\lambda]$ is triangular on matrix units, with diagonal entries equal to pairwise differences of eigenvalues of $\overline{D}_\lambda$. By (13.4), all those eigenvalues lie strictly in $(-1, 0)$, so every difference lies strictly in $(-1, 1)$. Equation (13.9) would make the positive integer d an eigenvalue of this commutator operator, which is impossible. Hence $d = 0$.

Therefore P is constant. Substitution into (13.5) gives $[C, P] = [D, P] = 0$, and every constant matrix commuting with C and D is plainly horizontal. Transposition gives the statement for $\mathcal{M}_q$. If a dual endomorphism acts on row vectors by right multiplication, the calculation

$$\nabla(\phi P) = (\nabla \phi)P \iff P' = [C + D/z, P]$$

shows that the same constancy result applies to the dual modules. $\square$

Corollary 13.4 (Multiplicity-free semisimplicity). *The horizontal endomorphism algebra of the moment connection is a commutative finite-dimensional $\mathbb{C}$ -algebra. In particular, the semisimple moment connection is multiplicity-free: its simple constituents are pairwise nonisomorphic.*

Proof. The matrix D has distinct eigenvalues, so it is diagonalizable and its centralizer is commutative. Apply Theorem 13.3. For a semisimple differential module, a repeated simple constituent contributes a full matrix algebra to the endomorphism algebra; commutativity therefore excludes repetitions. $\square$

Remark 13.5. The multiplicity-free conclusion is also visible geometrically. Monodromy at infinity is an n -cycle, so its cyclic subgroup acts simply transitively on a generic fiber; the restriction of the permutation representation to that subgroup is its regular representation and is multiplicity-free. If an irreducible constituent of the full monodromy representation occurred twice, then every cyclic character appearing in its restriction would occur at least twice, a contradiction.

14 Relation modules and constant projectors

Let

$$\Lambda = \{0, q(A), q(B)\}$$

with repetitions removed, and put

$$\mathcal{E} = \text{span}_{\mathbb{C}(z)} \{e^{\lambda z} : \lambda \in \Lambda\}. \quad (14.1)$$

If 0 is not an endpoint value, its forcing block is understood to be zero.

The functions

$$U_0, \dots, U_{n-2}, \{e^{\lambda z} : \lambda \in \Lambda\}$$

form a vector of E -functions: the endpoint exponents λ are algebraic, and $e^{\lambda z}$ is an E -function. Adjoining these exponential coordinates to (11.7) gives a homogeneous block system over $\overline{\mathbb{Q}}(z)$ with common denominator z . When 0 is not an endpoint value, the coordinate $e^{0z} = 1$ is adjoined with a zero forcing column. This coordinate will homogenize the value relation used below.

Define the relation module over $\mathbb{C}(z)$ by

$$\mathcal{R} = \{\phi \in \mathbb{C}(z)^{1 \times (n-1)} : \phi \mathbf{U} \in \mathcal{E}\}. \quad (14.2)$$

Lemma 14.1 (Relation subconnection). *The space $\mathcal{R}$ is a differential submodule of the dual homogeneous connection:*

$$\phi \in \mathcal{R} \implies \phi' + \phi \left(C + \frac{D}{z} \right) \in \mathcal{R}.$$

Proof. Let

$$A(z) = C + \frac{D}{z}, \quad \mathbf{f}(z) = \frac{1}{nz} \sum_{\lambda \in \Lambda} \mathbf{b}_\lambda e^{\lambda z},$$

where $\mathbf{b}_0 = 0$ when necessary. From (11.7),

$$(\phi \mathbf{U})' = (\phi' + \phi A) \mathbf{U} + \phi \mathbf{f}.$$

The left side belongs to $\mathcal{E}$ because $\mathcal{E}$ is stable under differentiation, and $\phi \mathbf{f} \in \mathcal{E}$. Hence

$$(\phi' + \phi A) \mathbf{U} \in \mathcal{E},$$

which is the required assertion. $\square$

Proposition 14.2 (A value relation forces an exponential polynomial). *If $U_0(1)$ is algebraic, then*

$$U_0(z) = \sum_{\lambda \in \Lambda} p_\lambda(z) e^{\lambda z} \quad (14.3)$$

for polynomials $p_\lambda \in \mathbb{C}[z]$.

Proof. Let $a = U_0(1) \in \overline{\mathbb{Q}}$. Order

$$\Lambda = \{0, \lambda_2, \dots, \lambda_s\}$$

and form the full column

$$\mathbf{F}(z) = (U_0(z), \dots, U_{n-2}(z), 1, e^{\lambda_2 z}, \dots, e^{\lambda_s z})^\top.$$

Put

$$B_\Lambda = \frac{1}{n} \begin{pmatrix} \mathbf{b}_0 & \mathbf{b}_{\lambda_2} & \cdots & \mathbf{b}_{\lambda_s} \end{pmatrix}, \quad \Delta_\Lambda = \text{diag}(0, \lambda_2, \dots, \lambda_s),$$

where $\mathbf{b}_0 = 0$ if $0 \notin \Lambda_\partial$. Then

$$\mathbf{F}' = \begin{pmatrix} C + D/z & B_\Lambda/z \\ 0 & \Delta_\Lambda \end{pmatrix} \mathbf{F}.$$

The constant coordinate $1 = e^{0z}$ makes $U_0(1) - a \cdot 1 = 0$ a homogeneous linear relation. More explicitly, let

$$\boldsymbol{\alpha} = (1, 0, \dots, 0; -a, 0, \dots, 0),$$

where the semicolon separates the moment and exponential coordinates. Then $\boldsymbol{\alpha}\mathbf{F}(1) = 0$. The block-system matrix has common denominator $T(z) = z$, and the point $\xi = 1$ satisfies $\xi T(\xi) = 1 \neq 0$. By Theorem 10.2, this prescribed linear form lifts to a polynomial coefficient row

$$R(z) \in \overline{\mathbb{Q}}[z]^{1 \times ((n-1)+s)}$$

such that

$$R(z)\mathbf{F}(z) = 0, \quad R(1) = \boldsymbol{\alpha}.$$

Writing

$$R(z) = (\boldsymbol{\phi}(z); c_0(z), c_{\lambda_2}(z), \dots, c_{\lambda_s}(z))$$

gives

$$\boldsymbol{\phi}(z)\mathbf{U}(z) + c_0(z) + \sum_{j=2}^s c_{\lambda_j}(z)e^{\lambda_j z} = 0, \quad \boldsymbol{\phi}(1) = \mathbf{e}_0 := (1, 0, \dots, 0).$$

Because $0 \in \Lambda$, the entire right-hand exponential combination belongs to $\mathcal{E}$; hence $\boldsymbol{\phi} \in \mathcal{R}$. We now regard this algebraic relation inside the complex differential module $\mathcal{R} \subseteq \mathbb{C}(z)^{n-1}$.

By Proposition 12.1, the homogeneous moment module and its dual are semisimple over $\mathbb{C}(z)$. The differential submodule $\mathcal{R}$ therefore has a horizontal complement and a horizontal idempotent projector. In row coordinates the projector acts by right multiplication by a rational matrix $P(z)$ with image $\mathcal{R}$. Horizontality is

$$P' = \left[C + \frac{D}{z}, P \right].$$

By Theorem 13.3, P is constant; substituting into the equation also gives $[C, P] = [D, P] = 0$. Let

$$W = \text{im}_{\mathbb{C}}(\cdot P) \subseteq \mathbb{C}^{1 \times (n-1)}.$$

Then $\mathcal{R} = W \otimes_{\mathbb{C}} \mathbb{C}(z)$. Since $\boldsymbol{\phi} = \boldsymbol{\phi}P$ and P is constant, evaluation at $z = 1$ gives

$$\mathbf{e}_0 = \boldsymbol{\phi}(1) = \boldsymbol{\phi}(1)P \in W.$$

Thus the constant row $\mathbf{e}_0$ lies in $\mathcal{R}$, and $U_0 = \mathbf{e}_0\mathbf{U} \in \mathcal{E}$.

It remains to upgrade rational exponential coefficients to polynomial ones. Choose a constant full-row-rank matrix S whose rows form a basis of W , and put $\mathbf{V} = S\mathbf{U}$. The space W is invariant under right multiplication by C and D : if $w = uP$, then

$$wC = uPC = uCP = (uC)P \in W, \quad wD = uPD = uDP = (uD)P \in W.$$

Hence there are unique constant matrices C_W, D_W such that

$$SC = C_W S, \quad SD = D_W S.$$

With $\mathbf{b}_{W,\lambda} = S\mathbf{b}_{\lambda}$, where $\mathbf{b}_0 = 0$ when $0 \notin \Lambda_{\partial}$, the moment system restricts to

$$\mathbf{V}' = \left(C_W + \frac{D_W}{z} \right) \mathbf{V} + \frac{1}{nz} \sum_{\lambda \in \Lambda} \mathbf{b}_{W,\lambda} e^{\lambda z}. \quad (14.4)$$

Every row of S belongs to $\mathcal{R}$, so every component of $\mathbf{V}$ lies in $\mathcal{E}$. By Lemma 10.3, there are unique rational vectors $\mathbf{r}_\lambda(z) \in \mathbb{C}(z)^{\dim W}$ such that

$$\mathbf{V}(z) = \sum_{\lambda \in \Lambda} \mathbf{r}_\lambda(z) e^{\lambda z}.$$

Equating the independent exponential coefficients in (14.4) gives

$$\mathbf{r}'_\lambda = (C_W - \lambda I) \mathbf{r}_\lambda + \frac{D_W}{z} \mathbf{r}_\lambda + \frac{1}{nz} \mathbf{b}_{W,\lambda}. \quad (14.5)$$

A pole of $\mathbf{r}_\lambda$ at $z_0 \neq 0$ is impossible: differentiation raises its order by one, whereas all terms on the right have at most the original order. At $z = 0$, suppose

$$\mathbf{r}_\lambda = z^{-m} v + O(z^{-m+1}), \quad m \geq 1, \quad v \neq 0.$$

The forcing term has pole order at most 1, strictly below $m + 1$ even when $m = 1$. Comparing the coefficient of z^{-m-1} therefore gives

$$-mv = D_W v.$$

Every eigenvalue of D_W is an eigenvalue of D . Indeed, if $u^\top D_W = \mu u^\top$, then full row rank of S gives $u^\top S \neq 0$, and

$$(u^\top S) D = u^\top D_W S = \mu (u^\top S).$$

By (11.8), $\text{Spec}(D_W) \subset (-1, 0)$, so it contains no negative integer $-m$. This contradiction excludes a pole at zero. Thus every rational vector $\mathbf{r}_\lambda$ has no finite pole and belongs to $\mathbb{C}[z]^{\dim W}$.

Finally choose a constant row a_0 with $\mathbf{e}_0 = a_0 S$. Then

$$U_0 = \mathbf{e}_0 \mathbf{U} = a_0 \mathbf{V} = \sum_{\lambda \in \Lambda} (a_0 \mathbf{r}_\lambda(z)) e^{\lambda z},$$

and $p_\lambda = a_0 \mathbf{r}_\lambda \in \mathbb{C}[z]$. In fact, the argument proves that every component of $\mathbf{V}$ is an exponential polynomial. $\square$

Remark 14.3. Entirety alone does not imply polynomial exponential coefficients:

$$\frac{e^z - 1}{z} = \frac{1}{z} e^z - \frac{1}{z}$$

is entire although both displayed coefficients have a pole. This is $U_0(z)$ for $q(x) = x$, $A = 0$, $B = 1$; that linear case has rank-zero moment connection and is handled separately by Lindemann–Weierstrass. For $n \geq 2$, the residue nonresonance in (11.8) is therefore load-bearing.

15 Constant-coefficient annihilation

Assume (14.3). Choose an integer N larger than the degrees of all nonzero polynomials p_λ ; equivalently, with the convention $\deg 0 = -\infty$, choose

$$N > \max_{\lambda \in \Lambda} \deg p_\lambda.$$

Define the constant-coefficient differential operator

$$L = \prod_{\lambda \in \Lambda} (\partial_z - \lambda)^N. \quad (15.1)$$

Then $LU_0 = 0$. Since differentiation under the integral sign multiplies by $q(x)$,

$$0 = LU_0(z) = \int_A^B H(q(x)) e^{zq(x)} dx, \quad (15.2)$$

where

$$H(t) = \prod_{\lambda \in \Lambda} (t - \lambda)^N \quad (15.3)$$

is nonzero. Because $0 \in \Lambda$ was introduced to homogenize the value relation, H includes a factor t^N even when 0 is neither an endpoint value nor a critical value; no geometric assertion about 0 is intended. Expanding at $z = 0$ gives

$$\int_A^B q(x)^m H(q(x)) dx = 0 \quad (m \geq 0). \quad (15.4)$$

We now rule out (15.4) for every nonconstant polynomial phase.

16 Phase-polynomial moment rigidity

Lemma 16.1 (Constellation representation). *Let $q \in \mathbb{C}[x]$ be nonconstant and let $A \neq B$. There exists a finite embedded tree T in the t -plane, containing all finite branch values of q and the endpoint values $q(A), q(B)$, together with an oriented path Γ from A to B in the finite graph $q^{-1}(T)$, such that on every open edge e of T the path is a signed finite union of analytic inverse branches $x_i(t)$ of q .*

Proof. Choose a regular center and pairwise disjoint analytic arcs from it to the distinct finite branch values and endpoint values, omitting repeated arms. Their union is an embedded star, hence a finite tree. Over the center, its preimage consists of the n sheets of the covering. The lifts of the branch arms join these sheets according to the branch-monodromy generators. Since the polynomial covering is connected, its monodromy action is transitive, so the resulting preimage graph is connected. Adding the endpoint arms preserves connectedness. The graph contains A and B , and therefore contains a path Γ joining them. On the interior of every edge the covering is unramified, so Γ decomposes there into signed analytic inverse branches. This is the standard polynomial constellation construction; see [6, §2.1]. $\square$

Lemma 16.2 (Phase-polynomial moment rigidity). *Let $q \in \mathbb{C}[x]$ be nonconstant, let $A \neq B$, and let $0 \neq H \in \mathbb{C}[t]$. Then the identities*

$$\int_A^B q(x)^m H(q(x)) dx = 0 \quad (m \geq 0)$$

cannot all hold.

Proof. Let T and Γ be supplied by Lemma 16.1. In particular, $q(\Gamma) \subseteq T$. Subdivide the finite graph $q^{-1}(T)$ at A and B , and regard Γ as an oriented finite edge path from A to B . For every $m \geq 0$, the differential

$$q(x)^m H(q(x)) dx$$

is a polynomial differential and hence has a polynomial primitive. Its integral depends only on the endpoints. Therefore the assumed identities remain valid on Γ :

$$\int_{\Gamma} q(x)^m H(q(x)) dx = 0 \quad (m \geq 0). \quad (16.1)$$

Define, for $s \in \mathbb{C} \setminus T$,

$$\mathcal{C}_H(s) = \int_{\Gamma} \frac{H(q(x))}{s - q(x)} dx. \quad (16.2)$$

The path Γ is compact and piecewise analytic, and the denominator in (16.2) is bounded away from zero locally uniformly for $s \in \mathbb{C} \setminus T$. Thus $\mathcal{C}_H$ is holomorphic on $\mathbb{C} \setminus T$, with differentiation under the integral sign justified on compact subsets. Put

$$M = \max_{x \in \Gamma} |q(x)|.$$

For $|s| > M$, the geometric series

$$\frac{1}{s - q(x)} = \sum_{m \geq 0} \frac{q(x)^m}{s^{m+1}}$$

converges uniformly on Γ . Using (16.1), we obtain

$$\mathcal{C}_H(s) = \sum_{m \geq 0} \frac{1}{s^{m+1}} \int_{\Gamma} q(x)^m H(q(x)) dx = 0 \quad (|s| > M).$$

A finite embedded tree has connected complement in the plane. Hence the identity theorem gives

$$\mathcal{C}_H \equiv 0 \quad \text{on } \mathbb{C} \setminus T. \quad (16.3)$$

We next express this Cauchy transform edge by edge. Orient every open edge e of T . Since all finite branch values of q are vertices of T , the map q is unramified over e . If $n = \deg q$, there are analytic inverse branches

$$x_{e,1}, \dots, x_{e,n} : e \longrightarrow \mathbb{C}, \quad q(x_{e,i}(t)) = t.$$

Let $f_{e,i} \in \mathbb{Z}$ be the signed number of times that Γ traverses the lift $x_{e,i}(e)$, relative to the chosen orientation of e , and set

$$\psi_e(t) = \sum_{i=1}^n f_{e,i} x'_{e,i}(t) = \sum_{i=1}^n \frac{f_{e,i}}{q'(x_{e,i}(t))}. \quad (16.4)$$

The edge integrals below are convergent even at branch-value vertices. Indeed, if c is an endpoint of e , $\xi \in q^{-1}(c)$ has ramification index $r \geq 1$, and a lifted branch tends to ξ , then locally

$$q(x) - c = (x - \xi)^r u(x), \quad u(\xi) \neq 0.$$

After choosing the corresponding branch of $(t - c)^{1/r}$ along e , one has

$$x_{e,i}(t) - \xi = O((t - c)^{1/r}), \quad x'_{e,i}(t) = O(|t - c|^{1/r-1}).$$

Since $1/r - 1 > -1$, this singularity is integrable. Changing variables $t = q(x)$ separately on every lifted edge therefore gives the absolutely convergent representation

$$\mathcal{C}_H(s) = \sum_e \int_e \frac{H(t) \psi_e(t)}{s - t} dt, \quad s \in \mathbb{C} \setminus T. \quad (16.5)$$

Fix an open edge e and an interior point $t_0 \in e$ with $H(t_0) \neq 0$. Choose a compact analytic subarc $\eta \subseteq e$ containing t_0 in its interior and a disk D about t_0 such that

$$\overline{D} \cap (T \setminus \eta) = \emptyset.$$

After shrinking η if necessary, the inverse branches occurring in (16.4) extend holomorphically to a neighborhood of η . Thus

$$\rho(t) := H(t)\psi_e(t)$$

extends holomorphically there. By the integrability just proved and the positive distance from $\overline{D}$ to $T \setminus \eta$, the function

$$R(s) := \mathcal{C}_H(s) - \int_{\eta} \frac{\rho(t)}{s-t} dt$$

extends holomorphically throughout D . Hence the entire lateral jump of $\mathcal{C}_H$ across η is the lateral jump of the local Cauchy integral

$$J(s) = \int_{\eta} \frac{\rho(t)}{s-t} dt.$$

We verify that jump directly. Let a, b be the initial and terminal endpoints of the oriented arc η . For s near t_0 , define

$$K(t, s) = \begin{cases} \frac{\rho(t) - \rho(s)}{t - s}, & t \neq s, \\ \rho'(s), & t = s. \end{cases}$$

The function K is jointly holomorphic near $\eta \times \eta$, and

$$J(s) = \rho(s) \int_{\eta} \frac{dt}{s-t} - \int_{\eta} K(t, s) dt. \quad (16.6)$$

The second term in (16.6) is holomorphic across η . On either side of the arc, the first scalar integral is a branch of

$$\int_{\eta} \frac{dt}{s-t} = \log(s-a) - \log(s-b).$$

Its two lateral determinations differ by $2\pi i \varepsilon_e$, where $\varepsilon_e \in \{\pm 1\}$ depends only on the orientation of η and on which side is denoted by $+$. Taking lateral limits in (16.6) therefore gives the local Sokhotski–Plemelj formula

$$(\mathcal{C}_H)_+(t_0) - (\mathcal{C}_H)_-(t_0) = 2\pi i \varepsilon_e H(t_0)\psi_e(t_0). \quad (16.7)$$

Both sides of the local arc lie in $\mathbb{C} \setminus T$, and (16.3) says that $\mathcal{C}_H$ vanishes identically on each side. The left side of (16.7) is therefore zero. Since $H(t_0) \neq 0$, we conclude that

$$\psi_e(t_0) = 0.$$

The point t_0 was arbitrary outside the finite zero set of the nonzero polynomial H . The function ψ_e is analytic on the open edge and hence continuous there, so it follows that

$$\psi_e \equiv 0 \quad \text{on every open edge } e \text{ of } T. \quad (16.8)$$

Finally, decompose the line integral of dx over the same lifted edges. The endpoint-integrability estimate above justifies the resulting improper edge integrals, and (16.8) gives

$$B - A = \int_{\Gamma} dx = \sum_e \sum_{i=1}^n f_{e,i} \int_e x'_{e,i}(t) dt = \sum_e \int_e \psi_e(t) dt = 0.$$

This contradicts $A \neq B$. Therefore the phase-moment identities cannot all hold. $\square$

17 Proof of the exponential-integral theorem

Proof of Theorem 9.1. Suppose first that $q(x) = ax + b$ with $a \neq 0$, and put

$$\alpha = aB + b, \quad \beta = aA + b.$$

Then $\alpha \neq \beta$ are algebraic and

$$\int_A^B e^{q(x)} dx = \frac{e^\alpha - e^\beta}{a}.$$

If this value were an algebraic number γ , then

$$e^\alpha - e^\beta - a\gamma e^0 = 0.$$

After combining terms if one of α, β equals 0, this is a nontrivial algebraic linear relation among exponentials of distinct algebraic numbers drawn from $\{\alpha, \beta, 0\}$. The Lindemann–Weierstrass theorem excludes such a relation; see [1, Chapter 1]. Hence the integral is transcendental.

Now let $n = \deg q \geq 2$. Apply the algebraic affine normalization of Equation (10.1); it suffices to treat the normalized polynomial, and we retain the notation q, A, B . Let

$$U_0(z) = \int_A^B e^{zq(x)} dx.$$

Assume, toward a contradiction, that $U_0(1)$ is algebraic. By Proposition 14.2, U_0 has the exponential-polynomial representation (14.3). Sections 15 and 16 then give a nonzero polynomial H satisfying the impossible moment identities (15.4). This contradiction proves that $U_0(1)$ is transcendental. $\square$

18 Completion of the proof of the Factorial Conjecture

Proof of Theorem 1.1. Assume that a nonzero counterexample exists. By Proposition 2.1, we may take $f \in \overline{\mathbb{Q}}[x, y]$. A nonzero constant has nonzero first factorial moment, so $D = \deg f \geq 1$.

If $A_0(t) = f_D(t, 1-t)$ is nonconstant, then Theorem 7.3 gives infinitely many nonzero factorial moments, a contradiction. Hence A_0 is constant, and Corollary 7.4 gives

$$f_D = a(x + y)^D, \quad a \in \overline{\mathbb{Q}}^\times.$$

Multiplying f by a^{-1} preserves the vanishing of all positive power moments, so we may assume $f_D = (x + y)^D$.

Put

$$q(t) = \frac{A_1(t)}{D} = \frac{f_{D-1}(t, 1-t)}{D} \in \overline{\mathbb{Q}}[t].$$

By Proposition 8.3,

$$\int_0^1 e^{q(t)} dt = 0. \tag{18.1}$$

If q is constant, the left side is the nonzero number e^q . If q is nonconstant, Theorem 9.1 with $A = 0$ and $B = 1$ says that the left side is transcendental and therefore nonzero. Both cases contradict (18.1). Thus no nonzero counterexample exists. $\square$

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